

Middle East Technical University

Department of Electrical and Electronics Engineering

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## EE568 – Design of Electrical Machines

### Project 2: Motor Winding Design & Analysis

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**Design:** Design 1:  $N_s = 15$ ,  $N_m = 8$

**Repository:** [github.com/seferidaci/EE568-Projects-sefer\\_idaci](https://github.com/seferidaci/EE568-Projects-sefer_idaci)

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# 1 Q1 – Integral-Slot Winding Design

## 1.1 Machine Parameters and Notation

The given machine has the following specifications:

Table 1: Given and derived parameters – Q1

| Parameter              | Meaning                            | Symbol | Value            |
|------------------------|------------------------------------|--------|------------------|
| Number of stator slots | Total slots around the stator bore | $N_s$  | 72               |
| Number of poles        | Total magnetic poles on the rotor  | $2p$   | 6                |
| Number of pole pairs   | Half the pole count; $p = 2p/2$    | $p$    | 3                |
| Number of phases       | Phases of the AC supply            | $m$    | 3                |
| Winding layers         | Number of coil sides per slot      | —      | 2 (double-layer) |

## 1.2 Step 1 – Derived Quantities

### 1.2.1 Electrical Angle per Slot $\alpha_e$

$$\boxed{\alpha_e = \frac{p \times 360^\circ}{N_s}} = \frac{3 \times 360^\circ}{72} = \mathbf{15^\circ} \text{ (electrical)} \quad (1)$$

### 1.2.2 Slots per Pole per Phase $q$

The total number of slots is shared equally among  $2p$  poles and  $m$  phases:

$$\boxed{q = \frac{N_s}{2p \cdot m}} = \frac{72}{6 \times 3} = \mathbf{4} \quad (2)$$

Because  $q = 4 \in \mathbb{Z}^+$ , this is an *integral-slot* winding. Each phase occupies 4 consecutive slots under each pole.

### 1.2.3 Pole Pitch $\tau_p$

The pole pitch is the number of slots under one pole:

$$\boxed{\tau_p = \frac{N_s}{2p}} = \frac{72}{6} = \mathbf{12} \text{ slots} \quad (3)$$

A *full-pitched* coil spans exactly  $\tau_p = 12$  slots.

### 1.2.4 One Pole-Pair = $360^\circ$ electrical = 24 slots

Since  $\alpha_e = 15^\circ$ , it takes  $360^\circ/15^\circ = 24$  slots to complete one full electrical cycle. This means **one pole-pair spans 24 consecutive slots**, which is the minimum repeating unit of the winding.

### 1.3 Step 2 – Upper Layer Assignment (the 60° Sector Rule)

#### 1.3.1 Physical Basis

Each stator slot has a conductor whose induced EMF is represented by a phasor. The phasor of slot  $k$  (upper layer) points at electrical angle:

$$\varphi_k = (k - 1) \alpha_e \pmod{360^\circ} \quad (4)$$

| Sector | Angle range              | Assignment |
|--------|--------------------------|------------|
| 1      | $[0^\circ, 60^\circ)$    | A+         |
| 2      | $[60^\circ, 120^\circ)$  | C–         |
| 3      | $[120^\circ, 180^\circ)$ | B+         |
| 4      | $[180^\circ, 240^\circ)$ | A–         |
| 5      | $[240^\circ, 300^\circ)$ | C+         |
| 6      | $[300^\circ, 360^\circ)$ | B–         |

#### 1.3.2 Slot Angle Table – One Pole-Pair (Slots 1–24)

Table 2: Upper-layer phase assignment for one pole-pair (72-slot, 6-pole)

| Slot $k$                | $\varphi_k$ | Sector | Layer-1 | Slot $k$                  | $\varphi_k$ | Sector | Layer-1 |
|-------------------------|-------------|--------|---------|---------------------------|-------------|--------|---------|
| 1                       | 0°          | 1      | A+      | 13                        | 180°        | 4      | A–      |
| 2                       | 15°         | 1      | A+      | 14                        | 195°        | 4      | A–      |
| 3                       | 30°         | 1      | A+      | 15                        | 210°        | 4      | A–      |
| 4                       | 45°         | 1      | A+      | 16                        | 225°        | 4      | A–      |
| 5                       | 60°         | 2      | C–      | 17                        | 240°        | 5      | C+      |
| 6                       | 75°         | 2      | C–      | 18                        | 255°        | 5      | C+      |
| 7                       | 90°         | 2      | C–      | 19                        | 270°        | 5      | C+      |
| 8                       | 105°        | 2      | C–      | 20                        | 285°        | 5      | C+      |
| 9                       | 120°        | 3      | B+      | 21                        | 300°        | 6      | B–      |
| 10                      | 135°        | 3      | B+      | 22                        | 315°        | 6      | B–      |
| 11                      | 150°        | 3      | B+      | 23                        | 330°        | 6      | B–      |
| 12                      | 165°        | 3      | B+      | 24                        | 345°        | 6      | B–      |
| <b>Pole 1 (0°–165°)</b> |             |        |         | <b>Pole 2 (180°–345°)</b> |             |        |         |

### 1.4 Step 3 – Lower Layer Assignment

#### 1.4.1 Full-Pitched Winding ( $y = 12$ )

Three representative slots are shown; all 24 follow the same logic:

| Slot $j$ | Layer-1 of slot $j - 12 \Rightarrow$ Layer-2 of slot $j$ |
|----------|----------------------------------------------------------|
| 1        | Layer-1[13] = A–      A+                                 |
| 5        | Layer-1[17] = C+      C–                                 |
| 13       | Layer-1[1] = A+      A–                                  |

| Slot:    | 1 | 2  | 3 | 4 | 5 | 6  | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|----------|---|----|---|---|---|----|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Layer 1: |   | A+ |   |   |   | C– |   |   |   | B+ |    |    |    | A– |    |    |    | C+ |    |    |    | B– |    |    |
| Layer 2: |   | A+ |   |   |   | C– |   |   |   | B+ |    |    |    | A– |    |    |    | C+ |    |    |    | B– |    |    |

The coil pitch arrow:  $\underbrace{1 \xrightarrow{y=12} 13}_{\text{Phase A coil 1}}, \underbrace{2 \rightarrow 14}_{\text{coil 2}}, \dots$

### 1.4.2 11/12 Short-Pitched Winding ( $y = 11$ )

Now  $y = 11$ : the angle difference between slot  $j$  and slot  $j - 11$  is  $11 \times 15^\circ = 165^\circ$  (not  $180^\circ$ ). At group boundaries the phasor falls in the adjacent sector instead of the opposite one, creating *mixed-phase* slots.

| Slot:    | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Layer 1: |    | A+ |    |    |    | C– |    |    |    | B+ |    |    |    | A– |    |    |    | C+ |    |    |    | B– |    |    |
| Layer 2: | A+ | A+ | A+ | C– | C– | C– | B+ | B+ | B+ | A– | A– | A– | C+ | C+ | C+ | B– | B– | B– | A+ | A+ | A+ | A+ | A+ | A+ |

The highlighted (yellow) slots (4, 8, 12, 16, 20, 24) are the *boundary slots* where two different phases share the same slot. They arise because shortening the pitch by 1 slot shifts the return conductor out of the purely opposite sector by  $180^\circ - 165^\circ = 15^\circ$  (one slot), crossing into the adjacent group.

## 1.5 Step 4 – Distribution Factor $k_d$

### 1.5.1 Physical Meaning

In a concentrated winding (all  $q$  coils of one phase under one pole placed in a single slot) the coil EMFs are perfectly in phase and add arithmetically. In a *distributed* winding the  $q$  coils are spread over  $q$  adjacent slots, each displaced by  $\alpha_e$  electrically. Their EMF phasors form an arc of a regular polygon, and the vector sum is smaller than the arithmetic sum.

$$k_d = \frac{\text{vector (phasor) sum of the } q \text{ coil EMFs}}{\text{arithmetic sum of the } q \text{ coil EMFs}} \quad (5)$$

### 1.5.2 Derivation

Each of the  $q$  phasors has magnitude  $E_m$  and successive phasors are offset by  $\alpha_e$ . The vector sum of  $q$  unit phasors starting at  $0^\circ$  and stepping by  $\alpha_e$ :

$$\left| \sum_{i=0}^{q-1} e^{j i \alpha_e} \right| = \frac{\sin(q \alpha_e / 2)}{\sin(\alpha_e / 2)} \quad (6)$$

(standard result for a finite geometric series on the unit circle). The arithmetic sum is simply  $q$ . Hence:

$$k_{d,n} = \frac{\sin\left(\frac{n q \alpha_e}{2}\right)}{q \sin\left(\frac{n \alpha_e}{2}\right)} \quad (7)$$

where  $n$  is the harmonic order. For the  $n$ -th harmonic the electrical angle between adjacent phasors becomes  $n \alpha_e$ , so  $\alpha_e$  is replaced by  $n \alpha_e$  throughout.

### 1.5.3 Calculations for $q = 4$ , $\alpha_e = 15^\circ$

**Fundamental ( $n = 1$ ):**

$$k_{d,1} = \frac{\sin\left(\frac{1 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{1 \times 15^\circ}{2}\right)} = \frac{\sin 30^\circ}{4 \sin 7.5^\circ} = \frac{0.5000}{4 \times 0.1305} = \frac{0.5000}{0.5221} = \mathbf{0.9577} \quad (8)$$

**Third harmonic ( $n = 3$ ):**

$$k_{d,3} = \frac{\sin\left(\frac{3 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{3 \times 15^\circ}{2}\right)} = \frac{\sin 90^\circ}{4 \sin 22.5^\circ} = \frac{1.0000}{4 \times 0.3827} = \frac{1.0000}{1.5307} = \mathbf{0.6533} \quad (9)$$

**Fifth harmonic ( $n = 5$ ):**

$$k_{d,5} = \frac{\sin\left(\frac{5 \times 4 \times 15^\circ}{2}\right)}{4 \sin\left(\frac{5 \times 15^\circ}{2}\right)} = \frac{\sin 150^\circ}{4 \sin 37.5^\circ} = \frac{0.5000}{4 \times 0.6088} = \frac{0.5000}{2.4351} = \mathbf{0.2053} \quad (10)$$

The distribution factor *always decreases with harmonic order* for a fixed  $q$  – distributing the winding inherently attenuates higher harmonics.

## 1.6 Step 5 – Pitch Factor $k_p$

### 1.6.1 Physical Meaning

Consider one coil with its go side in slot  $k$  and its return side in slot  $k + y$ . The EMF induced in the go-side conductor is  $e_{go}(t) = E_m \sin(\omega t)$ . The EMF induced in the return-side conductor is  $e_{ret}(t) = E_m \sin(\omega t - \theta_c)$  where  $\theta_c = y \alpha_e$  is the electrical angle spanned by the coil. Because the return conductor is wound backwards (it is the return path of the coil), its contribution to the *net* coil EMF is  $-e_{ret}$ . The total coil EMF is:

$$e_{coil} = e_{go} - e_{ret} = E_m [\sin \omega t - \sin(\omega t - \theta_c)] = 2E_m \cos\left(\omega t - \frac{\theta_c}{2}\right) \sin \frac{\theta_c}{2} \quad (11)$$

The amplitude is  $2E_m \sin(\theta_c/2)$ . For a *full-pitched* coil  $\theta_c = 180^\circ$ , so the amplitude is  $2E_m \sin(90^\circ) = 2E_m$  (maximum possible). The *pitch factor* is the ratio of the short-pitch amplitude to the full-pitch amplitude:

$$k_p = \frac{2E_m \sin(\theta_c/2)}{2E_m \sin(90^\circ)} = \sin \frac{\theta_c}{2} = \sin \frac{y \alpha_e}{2} = \sin \frac{y \pi}{2\tau_p} \quad (12)$$

For the  $n$ -th harmonic the field rotates  $n$  times faster, so the coil spans  $n$  times more electrical angle:

$$k_{p,n} = \sin \left( \frac{n y \pi}{2 \tau_p} \right) \quad (13)$$

### 1.6.2 Calculations

**Full-pitched** ( $y = 12, \tau_p = 12$ ):

$$k_{p,1} = \sin \frac{1 \times 12 \pi}{2 \times 12} = \sin \frac{\pi}{2} = \mathbf{1.0000} \quad (14)$$

$$k_{p,3} = \left| \sin \frac{3 \times 12 \pi}{2 \times 12} \right| = \left| \sin \frac{3\pi}{2} \right| = |-1| = \mathbf{1.0000} \quad (15)$$

$$k_{p,5} = \left| \sin \frac{5 \times 12 \pi}{2 \times 12} \right| = \left| \sin \frac{5\pi}{2} \right| = |+1| = \mathbf{1.0000} \quad (16)$$

All harmonics have  $k_p = 1$  – a full-pitched coil gives maximum EMF for every harmonic.

**11/12 Short-pitched** ( $y = 11, \tau_p = 12$ ):

$$k_{p,1} = \sin \frac{1 \times 11 \pi}{2 \times 12} = \sin \frac{11\pi}{24} = \sin 82.5^\circ = \mathbf{0.9914} \quad (17)$$

$$k_{p,3} = \left| \sin \frac{3 \times 11 \pi}{2 \times 12} \right| = \left| \sin \frac{33\pi}{24} \right| = |\sin 247.5^\circ| = |-\sin 67.5^\circ| = \mathbf{0.9239} \quad (18)$$

$$k_{p,5} = \left| \sin \frac{5 \times 11 \pi}{2 \times 12} \right| = \left| \sin \frac{55\pi}{24} \right| = |\sin 412.5^\circ| = \sin 52.5^\circ = \mathbf{0.7934} \quad (19)$$

## 1.7 Step 6 – Winding Factor $k_w$

The winding factor combines both effects:

$$k_{w,n} = k_{d,n} \times k_{p,n} \quad (20)$$

Table 3: Complete winding factor results – Full-pitched vs. 11/12 short-pitched

| $n$       | Full-pitched ( $y = 12$ ) |           |               | 11/12 Short-pitched ( $y = 11$ ) |           |               | Change       |            |
|-----------|---------------------------|-----------|---------------|----------------------------------|-----------|---------------|--------------|------------|
|           | $k_{d,n}$                 | $k_{p,n}$ | $k_{w,n}$     | $k_{d,n}$                        | $k_{p,n}$ | $k_{w,n}$     | $\Delta k_w$ | $\Delta\%$ |
| 1 (fund.) | 0.9577                    | 1.0000    | <b>0.9577</b> | 0.9577                           | 0.9914    | <b>0.9495</b> | −0.0082      | −0.9%      |
| 3         | 0.6533                    | 1.0000    | 0.6533        | 0.6533                           | 0.9239    | 0.6036        | −0.0497      | −7.6%      |
| 5         | 0.2053                    | 1.0000    | 0.2053        | 0.2053                           | 0.7934    | 0.1629        | −0.0424      | −20.6%     |

## 1.8 Step 7 – Winding Diagram (One Pole-Pair)

Table 4 shows the complete double-layer slot layout for one pole-pair (slots 1–24). The symbol  $\bullet$  means current flows *into* the page;  $\times$  means current flows *out of* the page. The coil pitch arrow  $k \rightarrow k + y$  connects each go-side (Layer 1) to its return-side (Layer 2).

Table 4: Winding diagram – full-pitched ( $y = 12$ ), one pole-pair

| Slot $k$  | Pole 1                                                                        |   |    |   |   |    |   |   |   |    |    |    | Pole 2 |    |    |    |    |    |    |    |    |    |    |    |  |
|-----------|-------------------------------------------------------------------------------|---|----|---|---|----|---|---|---|----|----|----|--------|----|----|----|----|----|----|----|----|----|----|----|--|
|           | 1                                                                             | 2 | 3  | 4 | 5 | 6  | 7 | 8 | 9 | 10 | 11 | 12 | 13     | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |  |
| Layer 1   |                                                                               |   | A● |   |   | C× |   |   |   |    | B● |    |        | A× |    |    |    | C● |    |    |    |    | B× |    |  |
| Layer 2   |                                                                               |   | A● |   |   | C× |   |   |   |    | B● |    |        | A× |    |    |    | C● |    |    |    |    | B× |    |  |
| Coil span | <div><div></div><div><math>y = 12 \text{ slots}</math></div><div></div></div> |   |    |   |   |    |   |   |   |    |    |    |        |    |    |    |    |    |    |    |    |    |    |    |  |

Table 5: Winding diagram – 11/12 short-pitched ( $y = 11$ ), one pole-pair. Yellow cells = boundary slots with mixed-phase layers.

| Slot $k$  | Pole 1                                                                                                                                                                                                                                                                                                                                          |    |    |    |    |    |    |    |    |    |    |    | Pole 2 |    |    |    |    |    |    |    |    |    |    |    |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|----|----|----|--------|----|----|----|----|----|----|----|----|----|----|----|
|           | 1                                                                                                                                                                                                                                                                                                                                               | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13     | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| Layer 1   |                                                                                                                                                                                                                                                                                                                                                 |    | A● |    |    | C× |    |    |    | B● |    |    |        | A× |    |    |    | C● |    |    |    | B× |    |    |
| Layer 2   | A●                                                                                                                                                                                                                                                                                                                                              | A● | A● | C× | C× | C× | C× | B● | B● | B● | B● | A× | A×     | A× | A× | C● | C● | C● | B× | B× | B× | B× | A● |    |
| Coil span | <div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div><div></div></div> <div><math>y = 11</math> slots</div> <div></div> |    |    |    |    |    |    |    |    |    |    |    |        |    |    |    |    |    |    |    |    |    |    |    |

## 1.9 Discussion and Comments

1. **Distribution factor decreases with harmonic order.** For  $q = 4$  and  $\alpha_e = 15^\circ$ , going from  $n = 1$  to  $n = 5$  drops  $k_d$  from 0.958 to only 0.205. Spreading the winding over  $q$  slots is therefore a powerful tool for harmonic attenuation, even before any pitch adjustment.
2. **Full-pitched winding: pitch factor is unity for all odd harmonics.** Because  $y\alpha_e = 180^\circ$ , the formula  $k_{p,n} = \sin(n\pi/2)$  gives  $\pm 1$  for every odd  $n$ . This means the full-pitched winding provides *no* pitch-based harmonic suppression – the 3rd harmonic winding factor of 0.653 is entirely due to  $k_d$  and remains high.
3. **Short-pitching disproportionately suppresses higher harmonics.** Shortening by just 1 slot ( $y = 11$ ) changes  $k_{p,1}$  by only  $-0.9\%$  but reduces  $k_{p,5}$  by  $-20.6\%$ . The mathematical reason:  $k_{p,n} = \sin(n \cdot y\pi/2\tau_p)$  is more sensitive to changes in  $y$  for larger  $n$  because the argument is multiplied by  $n$ . A small shift in  $y$  sweeps a larger angle for the 5th harmonic than for the fundamental.
4. **The 3rd harmonic is a triplen harmonic.** In a balanced 3-phase machine, triplen ( $n = 3, 9, 15, \dots$ ) space harmonics do not produce average torque because they are in phase across all three phases (no  $120^\circ$  offset between phases for  $n = 3$ ). However, they still cause iron losses and noise.
5. **Summary recommendation:** The 11/12 short-pitched winding is the better design. It costs  $< 1\%$  of useful torque capability while achieving significant attenuation of harmful harmonics. In industrial practice, short pitching is almost universally applied in distributed-winding AC machines.

## 2 Q2 – Fractional-Slot Winding Design

### 2.1 Machine Parameters and Notation

Table 6: Design 1 parameters ( $2575421 \bmod 5 = 1$ )

| Parameter                 | Symbol | Value |
|---------------------------|--------|-------|
| Number of stator slots    | $N_s$  | 15    |
| Number of poles (magnets) | $N_m$  | 8     |
| Number of pole pairs      | $p$    | 4     |
| Phases                    | $m$    | 3     |
| $R_{ro}/R_{so}$ ratio     | —      | 0.6   |

The slots per pole per phase is:

$$q = \frac{N_s}{N_m \cdot m} = \frac{15}{8 \times 3} = 0.625 \notin \mathbb{Z}^+ \quad (21)$$

Because  $q$  is not an integer, the simple coil-group rule of integral-slot windings (“ $q$  consecutive slots per phase per pole”) does not apply, and the closed-form formulas for  $k_d$  and  $k_p$  are invalid. Instead, the *star of slots* phasor method must be used.

### 2.2 Step 1 – Electrical Angle per Slot $\alpha_s$

For a machine with  $N_m$  poles the field repeats every pole pair ( $360^\circ$  electrical). The electrical angle between two adjacent slots is:

$$\alpha_s = \frac{N_m \times 180^\circ}{N_s} = \frac{8 \times 180^\circ}{15} = \mathbf{96^\circ} \quad (22)$$

Note that  $\alpha_s = 96^\circ > 60^\circ$ : each step jumps *across* a sector boundary rather than staying within one  $60^\circ$  phase group. This is the signature of a fractional-slot winding – the slot phasors do not cluster neatly into phase groups.

### 2.3 Step 2 – Slot Phasor Angles (Star of Slots)

For a fractional-slot winding the winding factor cannot be obtained from simple closed-form expressions; the *star of slots* (phasor diagram) method is used instead. Each slot  $k$  is assigned an EMF phasor whose electrical angle is:

$$\varphi_k = (k - 1) \alpha_s \pmod{360^\circ} \quad (23)$$

The  $N_s$  phasors so generated form the star of slots. The fundamental winding factor of each phase is, by definition, the magnitude of the phasor sum of its coil-sides divided by the number of coil-sides, so to maximise  $k_{w,1}$  each slot should be given to the phase whose axis its phasor most closely aligns with (with the sign chosen so that the contribution

adds rather than cancels). In practice this is implemented by the  $60^\circ$ -sector rule: the  $360^\circ$  circle is split into six  $60^\circ$  sectors labelled  $+A$ ,  $-C$ ,  $+B$ ,  $-A$ ,  $+C$ ,  $-B$ , and each slot is assigned to the phase/sign of the sector its phasor falls into – which is exactly the assignment that makes every coil-side contribute its maximum possible projection onto that phase's axis. Table 7 lists the calculation for all 15 slots. Because  $\gcd(15, 4) = 1$ , no two slots share the same angle – all 15 phasors are distinct and the sequence visits 15 different positions before returning to  $0^\circ$ .

Table 7: Slot phasor angles – Design 1 ( $\alpha_s = 96^\circ$ )

| Slot $k$ | $(k-1) \times 96^\circ$ | $\varphi_k$ | Sector | Slot $k$ | $(k-1) \times 96^\circ$ | $\varphi_k$ | Sector |
|----------|-------------------------|-------------|--------|----------|-------------------------|-------------|--------|
| 1        | 0                       | $0^\circ$   | 1      | 9        | 768                     | $48^\circ$  | 1      |
| 2        | 96                      | $96^\circ$  | 2      | 10       | 864                     | $144^\circ$ | 3      |
| 3        | 192                     | $192^\circ$ | 4      | 11       | 960                     | $240^\circ$ | 5      |
| 4        | 288                     | $288^\circ$ | 5      | 12       | 1056                    | $336^\circ$ | 6      |
| 5        | 384                     | $24^\circ$  | 1      | 13       | 1152                    | $72^\circ$  | 2      |
| 6        | 480                     | $120^\circ$ | 3      | 14       | 1248                    | $168^\circ$ | 3      |
| 7        | 576                     | $216^\circ$ | 4      | 15       | 1344                    | $264^\circ$ | 5      |
| 8        | 672                     | $312^\circ$ | 6      |          |                         |             |        |

## 2.4 Step 3 – Phase Assignment ( $60^\circ$ Sector Rule)

The same six-sector rule from Q1 (Section 1) applies here. The  $360^\circ$  circle is divided into six  $60^\circ$  sectors, and each slot is assigned to the phase whose sector contains its phasor:

| Sector | Angle range              | Assignment |
|--------|--------------------------|------------|
| 1      | $[0^\circ, 60^\circ)$    | A+         |
| 2      | $[60^\circ, 120^\circ)$  | C–         |
| 3      | $[120^\circ, 180^\circ)$ | B+         |
| 4      | $[180^\circ, 240^\circ)$ | A–         |
| 5      | $[240^\circ, 300^\circ)$ | C+         |
| 6      | $[300^\circ, 360^\circ)$ | B–         |

Applying this to Table 7 gives the full per-slot assignment in Table 8.

Table 8: Complete phase assignment – all 15 slots

| Slot | $\varphi_k$ | Sector | Assignment | Slot | $\varphi_k$ | Sector | Assignment |
|------|-------------|--------|------------|------|-------------|--------|------------|
| 1    | $0^\circ$   | 1      | A+         | 9    | $48^\circ$  | 1      | A+         |
| 2    | $96^\circ$  | 2      | C–         | 10   | $144^\circ$ | 3      | B+         |
| 3    | $192^\circ$ | 4      | A–         | 11   | $240^\circ$ | 5      | C+         |
| 4    | $288^\circ$ | 5      | C+         | 12   | $336^\circ$ | 6      | B–         |
| 5    | $24^\circ$  | 1      | A+         | 13   | $72^\circ$  | 2      | C–         |
| 6    | $120^\circ$ | 3      | B+         | 14   | $168^\circ$ | 3      | B+         |
| 7    | $216^\circ$ | 4      | A–         | 15   | $264^\circ$ | 5      | C+         |
| 8    | $312^\circ$ | 6      | B–         |      |             |        |            |

The result is balanced: each phase receives exactly  $15/3 = 5$  slots.

- **Phase A:** A+ at slots 1, 5, 9 and A– at slots 3, 7
- **Phase B:** B+ at slots 6, 10, 14 and B– at slots 8, 12
- **Phase C:** C+ at slots 4, 11, 15 and C– at slots 2, 13

Figure 1 shows all 15 slot phasors arranged in the star of slots, colour-coded by phase.

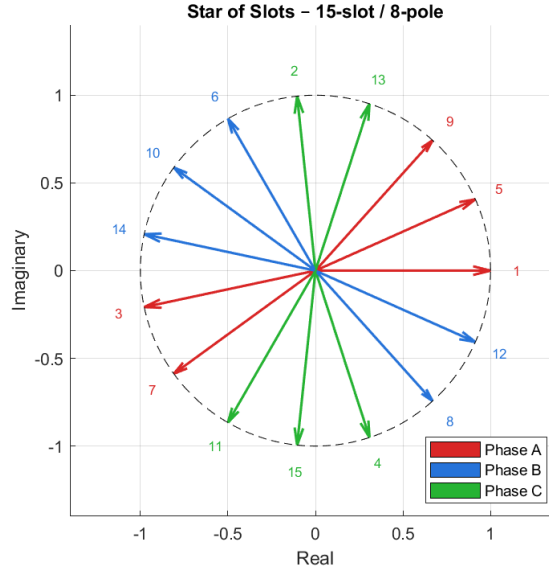


Figure 1: Star of slots for Design 1 ( $N_s = 15$ ,  $N_m = 8$ ). Colours: red = Phase A, blue = Phase B, green = Phase C. Solid arrows = positive direction; dashed = negative direction.

## 2.5 Step 4 – Winding Factor by Phasor Summation

### 2.5.1 Calculation Method

For the  $n$ -th space harmonic, each slot assigned to Phase A contributes one unit phasor. The general expression for the  $n$ -th harmonic winding factor by phasor summation is:

$$k_{w,n} = \frac{1}{N_{ph}} \left| \sum_{k \in \text{ph}^+} e^{jn\varphi_k} + \sum_{k \in \text{ph}^-} e^{j(n\varphi_k + \pi)} \right| \quad (24)$$

where  $N_{ph}$  is the number of slots per phase, and  $\text{ph}^+$ ,  $\text{ph}^-$  are the positive and negative slot sets of that phase. The angle of each contributing phasor therefore depends on whether the slot is positive or negative:

$$\text{A+ slot } k: e^{jn\varphi_k}, \quad \text{A- slot } k: e^{j(n\varphi_k + 180^\circ)} \quad (25)$$

The winding factor is then the magnitude of the vector sum divided by the number of slots per phase ( $N_{ph} = 5$ ):

$$k_{w,n} = \frac{|\Sigma_A|}{N_{ph}} = \frac{|\Sigma_A|}{5} \quad (26)$$

The A– slots effectively have their phasors *reversed* (rotated  $180^\circ$ ) before summing, which accounts for the fact that the current flows in the opposite direction in those slots.

### 2.5.2 Fundamental ( $n = 1$ )

Table 9: Phase A phasor contributions,  $n = 1$ 

| Slot                             | Type | $\varphi_k$ | Angle used ( $n\varphi_k + \text{rev.}$ ) | Real   | Imag.  |
|----------------------------------|------|-------------|-------------------------------------------|--------|--------|
| 1                                | A+   | $0^\circ$   | $0^\circ$                                 | +1.000 | +0.000 |
| 5                                | A+   | $24^\circ$  | $24^\circ$                                | +0.914 | +0.407 |
| 9                                | A+   | $48^\circ$  | $48^\circ$                                | +0.669 | +0.743 |
| 3                                | A–   | $192^\circ$ | $192^\circ + 180^\circ = 12^\circ$        | +0.978 | +0.208 |
| 7                                | A–   | $216^\circ$ | $216^\circ + 180^\circ = 36^\circ$        | +0.809 | +0.588 |
| <b>Sum <math>\Sigma_A</math></b> |      |             |                                           | +4.370 | +1.946 |

$$|\Sigma_A| = \sqrt{4.370^2 + 1.946^2} = \sqrt{19.10 + 3.79} = 4.783 \quad (27)$$

$$k_{w,1} = \frac{4.783}{5} = 0.9566 \quad (28)$$

**Observation:** After reversing the A– phasors (slots 3 and 7), all five phasors cluster in the first quadrant within a  $48^\circ$  arc. They nearly align, which explains the high winding factor.

### 2.5.3 Third Harmonic ( $n = 3$ )

For  $n = 3$  the phasor angle of each slot is multiplied by 3 before applying the reversal for A– slots.

Table 10: Phase A phasor contributions,  $n = 3$ 

| Slot                             | Type | $\varphi_k$ | Angle used ( $3\varphi_k + \text{rev.}$ ) | Real   | Imag.  |
|----------------------------------|------|-------------|-------------------------------------------|--------|--------|
| 1                                | A+   | $0^\circ$   | $0^\circ$                                 | +1.000 | +0.000 |
| 5                                | A+   | $24^\circ$  | $72^\circ$                                | +0.309 | +0.951 |
| 9                                | A+   | $48^\circ$  | $144^\circ$                               | –0.809 | +0.588 |
| 3                                | A–   | $192^\circ$ | $576^\circ + 180^\circ = 36^\circ$        | +0.809 | +0.588 |
| 7                                | A–   | $216^\circ$ | $648^\circ + 180^\circ = 108^\circ$       | –0.309 | +0.951 |
| <b>Sum <math>\Sigma_A</math></b> |      |             |                                           | +1.000 | +3.078 |

$$|\Sigma_A| = \sqrt{1.000^2 + 3.078^2} = \sqrt{1.000 + 9.474} = 3.236 \quad (29)$$

$$k_{w,3} = \frac{3.236}{5} = 0.6472 \quad (30)$$

### 2.5.4 Fifth Harmonic ( $n = 5$ )

Table 11: Phase A phasor contributions,  $n = 5$ 

| Slot           | Type | $\varphi_k$ | Angle used ( $5\varphi_k + \text{rev.}$ ) | Real   | Imag.  |
|----------------|------|-------------|-------------------------------------------|--------|--------|
| 1              | A+   | $0^\circ$   | $0^\circ$                                 | +1.000 | +0.000 |
| 5              | A+   | $24^\circ$  | $120^\circ$                               | −0.500 | +0.866 |
| 9              | A+   | $48^\circ$  | $240^\circ$                               | −0.500 | −0.866 |
| 3              | A−   | $192^\circ$ | $960^\circ + 180^\circ = 60^\circ$        | +0.500 | +0.866 |
| 7              | A−   | $216^\circ$ | $1080^\circ + 180^\circ = 180^\circ$      | −1.000 | +0.000 |
| Sum $\Sigma_A$ |      |             |                                           | −0.500 | +0.866 |

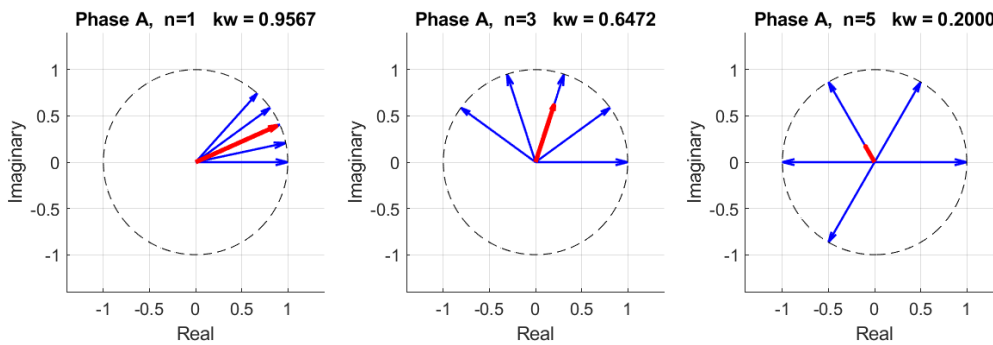
$$|\Sigma_A| = \sqrt{(-0.500)^2 + (0.866)^2} = \sqrt{0.250 + 0.750} = 1.000 \quad (31)$$

$$k_{w,5} = \frac{1.000}{5} = 0.200 \quad (32)$$

Table 12: Summary of winding factors – Design 1 (15-slot/8-pole)

| Harmonic          | $\Sigma_A$        | $ \Sigma_A $ | $k_{w,n}$     | Note                             |
|-------------------|-------------------|--------------|---------------|----------------------------------|
| $n = 1$ (fund.)   | $4.370 + j1.946$  | 4.783        | <b>0.9566</b> | High, good winding configuration |
| $n = 3$ (triplen) | $1.000 + j3.078$  | 3.236        | 0.6472        | Zero-seq. no net torque          |
| $n = 5$           | $-0.500 + j0.866$ | 1.000        | 0.2000        | Suppressed                       |

Figure 2 shows the Phase A phasor diagrams for all three harmonics. Each arrow is one slot phasor (after reversing A− slots); the thick red arrow is the vector resultant.

Figure 2: Phase A phasor diagrams for  $n = 1, 3, 5$ .

### Note on the Concentrated-Coil Convention

The phasor sum above uses only the five *go-side* positions of Phase A (Layer 1: slots 1, 5, 9, 3, 7). The return-sides (Layer 2) of these coils are *not* added as extra phasors – and this is intentional.

In a fractional-slot **concentrated** (single-tooth) winding each coil is wound around *one* tooth. Following the convention used in the Hanselman book, each coil is represented by a *single phasor placed at its tooth-centre angle*, and the winding factor is the vector sum of the five coil phasors divided by 5:

$$k_{w,n} = \frac{|\sum_{i=1}^5 e^{jn\varphi_i}|}{5} \quad (\text{coil phasors at tooth centres}). \quad (33)$$

## 2.6 Step 5 – Constructing the Winding Diagram

The star of slots provides one phase assignment per slot, which is taken as the upper layer (Layer 1) of each slot. In this concentrated tooth-coil topology each coil is wound around a single tooth, so its go-side sits in Layer 1 of slot  $k$  and its return-side in Layer 2 of slot  $k + 1$ . Because the return-side carries the opposite current direction, the lower layer of any slot is simply the negative of the upper layer of the preceding slot.

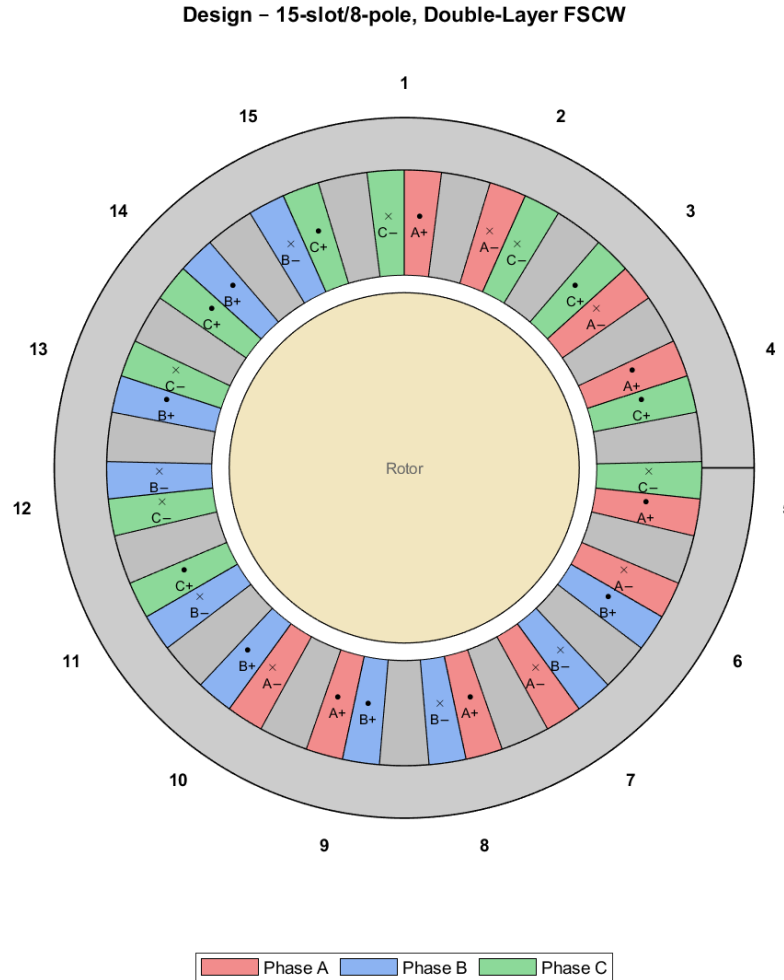


Figure 3: Winding layout for Design 1 (15-slot/8-pole, 3-phase).

## 2.7 Discussion

1. **High fundamental winding factor for a concentrated winding.**  $k_{w,1} = 0.9566$  is comparable to that of distributed windings. The 15/8 combination is identified in the Hanselman book as one of the best fractional-slot combinations for torque density, and the value  $k_{w,1} = 0.9566$  obtained here matches the example tabulated in the book.
2. **Natural suppression of the 5th harmonic.** The three A+ slots contribute phasors at  $0^\circ$ ,  $120^\circ$ ,  $240^\circ$  for  $n = 5$  – a balanced triphasic set that sums to zero. Only the A– slots restore a small residual ( $|\Sigma| = 1.0$ ), giving  $k_{w,5} = 0.2$ . This suppression is geometric and requires no pitch adjustment.
3. **3rd harmonic: large winding factor, zero net rotating field.**  $k_{w,3} = 0.647$  is large, but as discussed in Q1, the 3rd space harmonic is triplen: balanced condition  $I_A + I_B + I_C = 0$  cancels the net rotating component. It may still cause iron losses through standing-wave effects.
4. **Two phases share each slot.** Unlike integral-slot windings where each slot is occupied by a single phase, the double-layer FSCW places two *different* phases in each slot. Thicker Insulation is needed.
5. **Short end-windings – key FSCW advantage.** Because each coil spans only one tooth pitch (approximately  $360^\circ/(N_s) = 24^\circ$  mechanical), the overhang is very short. This reduces copper volume, DC resistance, and copper losses.

## 3 Q3 – Analytical Modelling

### 3.1 Machine Dimensions

From the assignment common specifications and the Design 1 ratio  $R_{ro}/R_{so} = 0.6$ . This split between rotor and stator is not a free design parameter; the Common Specs require that “the rotor outside radius  $R_m$  is chosen to maximise the motor constant  $K_m = T/\sqrt{P_{Cu}}$ .” For surface-PM machines the motor constant can be written  $K_m \propto B_g \alpha_m R_{ro} \sqrt{A_{slot}}$ , where the slot area itself depends on the radial budget left over for the slots,  $R_{so} - (R_{ro} + g + w_{sy})$ . Differentiating with respect to  $R_{ro}$  and equating to zero gives an optimum split that, for the assumed loadings and a magnet thickness of  $l_m = 4$  mm, falls near  $R_{ro}/R_{so} \approx 0.55$ – $0.65$ ; Hanselman tabulates the precise value of 0.60 for Design 1 of the 15-slot/8-pole family. The geometry inputs that follow take this as given:

Table 13: Machine geometry – Design 1

| Parameter                    | Symbol                    | Value               |
|------------------------------|---------------------------|---------------------|
| Stator outer radius          | $R_{so}$                  | 50.0 mm             |
| Rotor outer radius           | $R_{ro} = 0.6 R_{so}$     | 30.0 mm             |
| Air-gap length               | $g$                       | 1.0 mm              |
| Magnet radial thickness      | $l_m$                     | 4.0 mm              |
| Stator inner radius          | $R_{si} = R_{ro} + g$     | 31.0 mm             |
| Rotor core outer radius      | $R_{ro,c} = R_{ro} - l_m$ | 26.0 mm             |
| Stack length                 | $L_s$                     | 100 mm              |
| Magnet remanence             | $B_r$                     | 1.3 T               |
| Magnet relative permeability | $\mu_r$                   | 1.05                |
| Magnet arc ratio             | $\alpha_m$                | 0.889               |
| Target iron flux density     | $B_t$                     | 1.4 T               |
| Current density (rms)        | $J_{rms}$                 | 5 A/mm <sup>2</sup> |
| Slot fill factor             | $k_{wc}$                  | 0.60                |

### 3.2 Slot Pitch and Tooth Geometry

The slot pitch  $\tau_s$  is the circumferential distance between the centrelines of two adjacent slots, measured at the stator bore:

$$\tau_s = \frac{2\pi R_{si}}{N_s} = \frac{2\pi \times 31}{15} = 12.99 \text{ mm} \quad (34)$$

where  $R_{si}$  is the stator inner radius and  $N_s$  the total slot count. Figure 4 shows the relevant geometry in a linearised (developed) view of one slot pitch.

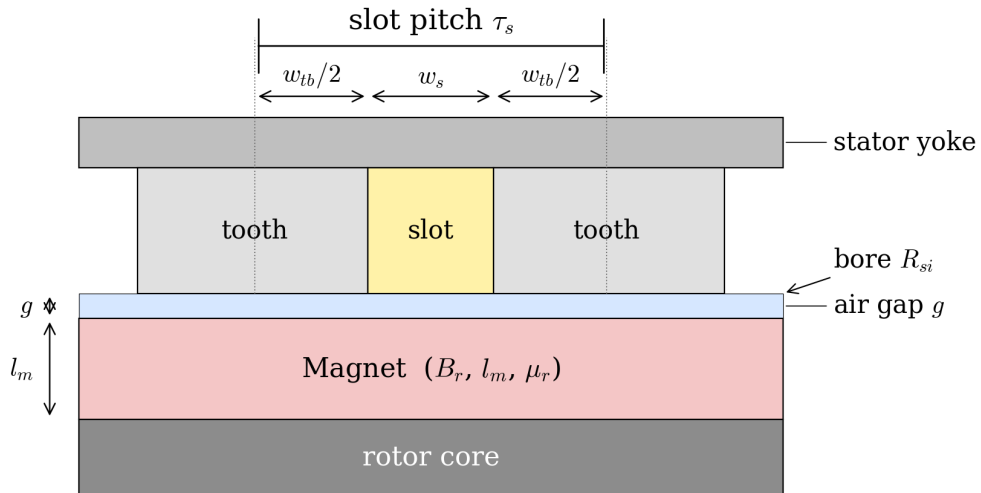


Figure 4: Developed (unrolled) view of the stator bore showing one slot pitch  $\tau_s$ , the tooth body width  $w_{tb}$  (split half-and-half between the two neighbouring teeth), the slot opening  $w_s$ , the air gap  $g$ , and the surface magnet of thickness  $l_m$ .

A first-pass estimate of the peak air-gap flux density, ignoring the slotting effect ( $k_c =$

1), follows from a one-dimensional magnetic equivalent circuit of a surface-PM rotor with a smooth stator. Assuming the stator iron is infinitely permeable (no  $H$ -drop in the steel) and that the magnet and air-gap cross-sections are equal so that  $B_m = B_g$  by flux continuity, two relations govern the operating point:

- Magnet B–H line (linearised recoil curve):  $B_m = B_r + \mu_0 \mu_r H_m \implies H_m = (B_m - B_r)/(\mu_0 \mu_r)$ .
- Ampère’s law around one pole loop (magnet  $\rightarrow$  gap  $\rightarrow$  iron  $\rightarrow$  gap  $\rightarrow$  magnet, iron contributes nothing):  $H_m l_m + H_g g = 0$ , with  $H_g = B_g/\mu_0$ .

Substituting the B–H line into Ampère’s law and using  $B_m = B_g$ :

$$\frac{B_g - B_r}{\mu_0 \mu_r} l_m = -\frac{B_g}{\mu_0} g \quad (35)$$

Multiplying through by  $\mu_0$ , expanding, and collecting the  $B_g$  terms gives  $B_g (l_m/\mu_r + g) = B_r l_m/\mu_r$ , which rearranges to:

$$B_{g,0} = \frac{B_r l_m}{l_m + \mu_r g} = \frac{1.3 \times 4}{4 + 1.05 \times 1} = 1.030 \text{ T} \quad (36)$$

Equivalently, the magnet behaves as a flux source  $B_r$  in series with its own reluctance  $\mathcal{R}_m = l_m/(\mu_0 \mu_r A)$ , loaded by the gap reluctance  $\mathcal{R}_g = g/(\mu_0 A)$ , and  $B_g/B_r$  is the reluctance-divider ratio  $\mathcal{R}_m/(\mathcal{R}_m + \mathcal{R}_g) = l_m/(l_m + \mu_r g)$ . Figure 5 shows the corresponding magnetic equivalent circuit. With Carter’s coefficient included (next subsection),  $g$  is replaced by the effective gap  $k_c g$ , giving the corrected  $B_{g,\text{peak}} = B_r l_m/(l_m + \mu_r k_c g)$  used below.

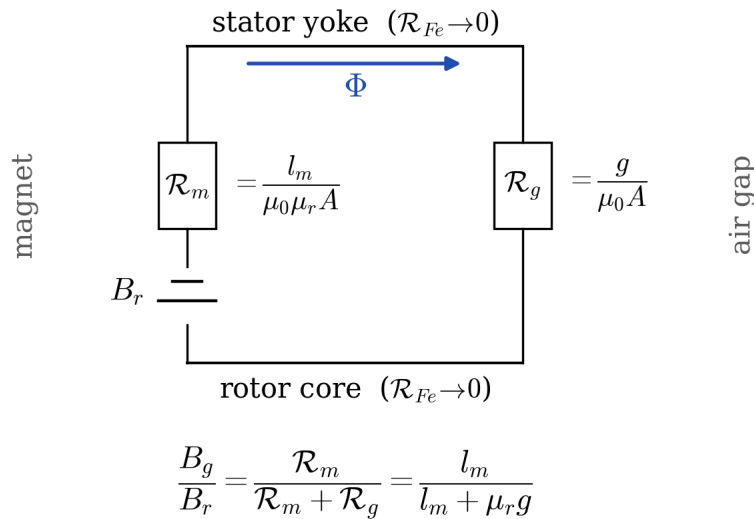


Figure 5: One-dimensional magnetic equivalent circuit of the surface-PM rotor. The magnet is represented by the MMF source  $B_r$  in series with its reluctance  $\mathcal{R}_m$ ; the air gap by  $\mathcal{R}_g$ ; the iron pieces by ideal shorts.  $B_g/B_r$  is the reluctance-divider ratio  $\mathcal{R}_m/(\mathcal{R}_m + \mathcal{R}_g) = l_m/(l_m + \mu_r g)$ .

The corresponding average flux density seen by one tooth – the magnet arc spans only a fraction  $\alpha_m$  of the pole pitch – is:

$$B_{g,\text{avg}} = B_{g,0} \alpha_m = 1.030 \times 0.889 = 0.916 \text{ T} \quad (37)$$

Conservation of flux requires the flux entering the air gap over one slot pitch to be funnelled through one tooth of width  $w_{tb}$  (Figure 6). Setting the tooth flux density equal to the target value  $B_t$  gives:

$$B_{g,\text{avg}} \cdot \tau_s = B_t \cdot w_{tb} \implies w_{tb} = \frac{B_{g,\text{avg}} \tau_s}{B_t} = \frac{0.916 \times 12.99}{1.4} = 8.50 \text{ mm} \quad (38)$$

The remaining width of the slot pitch is taken as the slot opening:

$$w_s = \tau_s - w_{tb} = 12.99 - 8.50 = 4.49 \text{ mm} \quad (39)$$

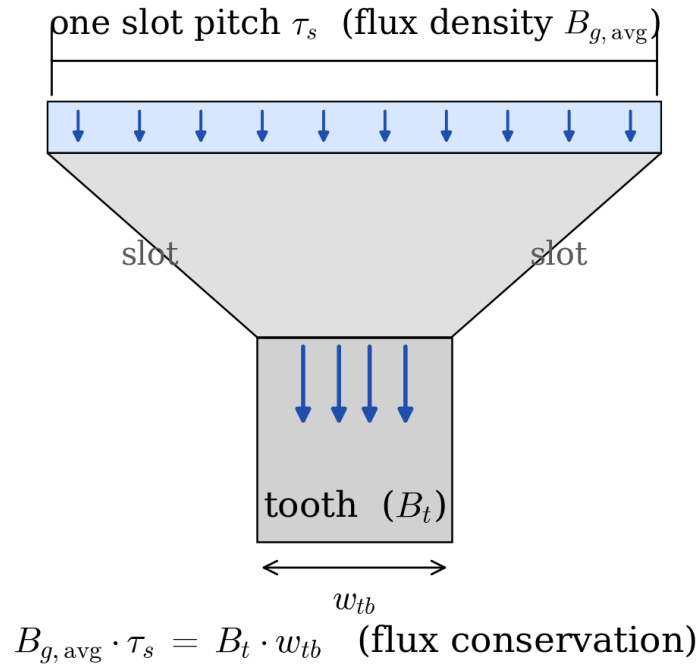


Figure 6: Flux funnelling. The air-gap flux entering the stator over one slot pitch  $\tau_s$  at average density  $B_{g,\text{avg}}$

### 3.3 Carter's Coefficient

Stator slotting causes the air-gap flux density to dip above each slot opening, which raises the average reluctance of the gap. Carter's coefficient  $k_c \geq 1$  captures this effect by replacing the slotted stator with an equivalent smooth surface having an enlarged air gap  $k_c g$ .

$$k_c = \frac{\tau_s}{\tau_s - \kappa w_s}, \quad \kappa = \frac{w_s/g}{5 + w_s/g} \quad (40)$$

The symbols are:

- $\tau_s = 2\pi R_{si}/N_s$  – slot pitch
- $w_s$  – slot opening width

- $g$  – air-gap length;
- $\kappa$  – dimensionless slot-opening factor

Applying Eq. (40) with  $g = 1$  mm and  $w_s = 4.49$  mm:

$$\frac{w_s}{g} = 4.49, \quad \kappa = \frac{4.49}{5 + 4.49} = \frac{4.49}{9.49} = 0.473 \quad (41)$$

$$k_c = \frac{12.99}{12.99 - 0.473 \times 4.49} = \frac{12.99}{12.99 - 2.124} = \frac{12.99}{10.866} = \mathbf{1.195} \quad (42)$$

### 3.4 Effective Air Gap and Corrected Flux Density

For a surface-mount PM machine with radially magnetised magnets the rotor surface and stator bore are treated as concentric cylinders, and the peak air-gap flux density is:

$$B_{g,\text{peak}} = \frac{B_r l_m}{l_m + \mu_r k_c g} \quad (43)$$

where  $B_r$  is the magnet remanence,  $l_m$  the magnet radial thickness,  $\mu_r$  the magnet relative permeability,  $g$  the physical air gap, and  $k_c$  Carter's coefficient. The effective air gap including the magnet contribution is:

$$g_{\text{eff}} = k_c g + \frac{l_m}{\mu_r} = 1.195 \times 1 + \frac{4}{1.05} = 1.195 + 3.810 = 5.005 \text{ mm} \quad (44)$$

Substituting numerical values:

$$B_{g,\text{peak}} = \frac{1.3 \times 4}{4 + 1.05 \times 1.195 \times 1} = \frac{5.2}{5.255} = \mathbf{0.989 \text{ T}} \quad (45)$$

### 3.5 Effective Axial Length

Fringing at the axial ends of the machine extends the flux-linking length beyond the physical stack length  $L_s$ . A common approximation is:

$$L_e = L_s + 2g \quad (46)$$

Numerically:

$$L_e = 100 + 2 \times 1 = \mathbf{102 \text{ mm}} \quad (47)$$

### 3.6 Air-Gap Peak Flux Density Summary

All three values are consistent: the slotting correction reduces  $B_{g,\text{peak}}$  from 1.030 T (idealised) to 0.989 T (with Carter correction), a reduction of 3.9%.

### 3.7 Flux Per Pole

The flux per pole, needed to size the back-iron and to calculate the induced EMF, is:

$$\Phi_{pole} = B_{g,peak} \alpha_m \tau_p L_e, \quad \tau_p = \frac{\pi R_{si}}{p} \quad (48)$$

where  $\alpha_m = \theta_m/180^\circ$  is the magnet arc ratio ( $160^\circ/180^\circ = 0.889$  here) and  $\tau_p$  is the pole pitch at the stator bore. Evaluating:

$$\tau_p = \frac{\pi \times 31}{4} = 24.35 \text{ mm} \quad (49)$$

$$\Phi_{pole} = 0.989 \times 0.889 \times 24.35 \times 10^{-3} \times 102 \times 10^{-3} = \mathbf{2.18 \text{ mWb}} \quad (50)$$

### 3.8 Magnetic Loading

The magnetic loading  $\bar{B}$  is the average value of the radial air-gap flux density taken over one full pole pitch. For a surface-mount PM machine with a magnet arc ratio  $\alpha_m$ , the flux is essentially uniform under the magnet and (for the analytical model) zero between magnets, so:

$$\bar{B} = B_{g,peak} \cdot \alpha_m = 0.989 \times 0.889 = \mathbf{0.879 \text{ T}} \quad (51)$$

$\bar{B}$  together with the electrical loading  $A_{rms}$  determine the shear stress (and hence the torque) the machine can develop, as expressed by Eq. (63) below.

### 3.9 Iron Yoke Widths

The stator and rotor yoke widths are sized so that the peak flux density in the iron does not exceed the target  $B_t = 1.4 \text{ T}$ :

$$w_{sy} = w_{ry} = \frac{\Phi_{pole}}{2 B_t L_e} \quad (52)$$

Numerically:

$$w_{sy} = w_{ry} = \frac{2.18 \times 10^{-3}}{2 \times 1.4 \times 0.102} = \mathbf{7.64 \text{ mm}} \quad (53)$$

The rotor shaft radius is  $R_{ro,c} - w_{ry} = 26 - 7.64 = 18.4 \text{ mm}$ .

#### Self-consistency check on tooth and yoke flux density

The tooth and yoke widths above were sized using the pre-Carter estimate  $B_{g,0} = 1.030 \text{ T}$ . After the Carter correction the actual peak air-gap flux density drops to  $0.989 \text{ T}$ , so the iron flux densities should be re-checked. By flux conservation, the actual tooth flux density is

$$B_{tooth}^{actual} = \frac{B_{g,peak} \alpha_m \tau_s}{w_{tb}} = \frac{0.989 \times 0.889 \times 12.99}{8.50} \approx \mathbf{1.343 \text{ T}} \quad (54)$$

which sits just below the  $B_t = 1.4\text{ T}$  target – the slotting correction provides a small natural margin in the teeth. The yoke flux density, found from the actual pole flux  $\Phi_{pole} = 2.18\text{ mWb}$ , is

$$B_{\text{yoke}}^{\text{actual}} = \frac{\Phi_{pole}}{2 w_{sy} L_e} = \frac{2.18 \times 10^{-3}}{2 \times 7.64 \times 10^{-3} \times 0.102} \approx \mathbf{1.398\text{ T}} \quad (55)$$

which lands essentially at the design target (this is by construction, since  $w_{sy}$  was sized for  $B_t = 1.4\text{ T}$  with the Carter-corrected  $\Phi_{pole}$ ). The two cross-checks confirm the analytical design is self-consistent at the chosen  $1.4\text{ T}$  limit.

### 3.10 Electrical Loading

Available slot height and slot area (rectangular approximation):

$$h_{slot} = (R_{so} - R_{si}) - w_{sy} = 19 - 7.64 = 11.36\text{ mm} \quad (56)$$

$$A_{slot} = w_s \times h_{slot} = 4.49 \times 11.36 = 51.0\text{ mm}^2 \quad (57)$$

The linear current density (electrical loading) characterises the ampere-conductor loading on the stator bore circumference:

$$A_{rms} = \frac{J_{rms} k_{wc} A_{slot}}{\tau_s} \quad (58)$$

where  $J_{rms}$  is the rms current density in the conductors,  $k_{wc}$  the slot fill factor, and  $A_{slot}$  the net slot cross-section. Substituting:

$$A_{rms} = \frac{5 \times 0.60 \times 51.0}{12.99} = \mathbf{11.78\text{ kA/m}} \quad (59)$$

#### Phase current, wire size and slot-fill verification

Anticipating  $N_{cs} = 8$  conductors per slot from Section 3.13, the total ampere-conductors per slot is  $N_{cs} I_{rms} = A_{rms} \tau_s$ , so the phase RMS current is

$$I_{rms} = \frac{A_{rms} \tau_s}{N_{cs}} = \frac{11.78 \times 10^3 \times 12.99 \times 10^{-3}}{8} = \mathbf{19.1\text{ A}} \quad (60)$$

The required net copper area per conductor follows from the prescribed current density  $J_{rms} = 5\text{ A/mm}^2$ :

$$A_{\text{wire}} = \frac{I_{rms}}{J_{rms}} = \frac{19.1}{5} = \mathbf{3.83\text{ mm}^2} \quad (61)$$

The actual slot fill is then

$$k_{wc}^{\text{check}} = \frac{N_{cs} A_{\text{wire}}}{A_{slot}} = \frac{8 \times 3.83}{51.0} = 0.601 \quad (62)$$

which matches the assumed 0.60 by construction (since  $A_{rms}$  was derived from  $J_{rms}$  and  $k_{wc}$ ); the check confirms that 8 conductors of the calculated wire area fit the slot without violating the fill-factor target. Since it is fractional slot there may be 2 different phases 4 wires for each.

### 3.11 Shear Stress and Expected Torque

Tangential force per unit air-gap surface area arises from the interaction between the radial flux density produced by the magnets and the tangential current sheet produced by the stator winding. Time-averaging over one electrical cycle, the electromagnetic shear stress at the stator bore is:

$$\sigma = \frac{1}{2} k_{w,1} B_{g,\text{peak}} A_{\text{peak}} = \frac{1}{\sqrt{2}} k_{w,1} B_{g,\text{peak}} A_{\text{rms}} \quad (63)$$

The fundamental winding factor  $k_{w,1}$  accounts for the fact that only the fundamental space harmonic of the winding produces useful torque; the factor of  $1/\sqrt{2}$  converts from peak to rms current loading. Integrating  $\sigma$  over the stator bore area  $2\pi R_{si} L_e$  and multiplying by the lever arm  $R_{si}$  gives the average electromagnetic torque, and multiplying by the mechanical angular speed  $\omega_m = 2\pi n/60$  gives the shaft power:

$$T = \sigma \cdot 2\pi R_{si}^2 L_e, \quad P = T \omega_m \quad (64)$$

Inserting the values computed above:

$$\sigma = \frac{1}{\sqrt{2}} \times 0.9566 \times 0.989 \times 11,780 = \mathbf{7.87 \text{ kPa}} \quad (65)$$

$$T = 7870 \times 2\pi \times (0.031)^2 \times 0.102 = \mathbf{4.85 \text{ N} \cdot \text{m}} \quad (66)$$

$$P = 4.85 \times \frac{1500 \times 2\pi}{60} = \mathbf{762 \text{ W}} \quad (67)$$

### 3.12 Specific Machine Constant

The classical output equation collects the magnetic loading  $\hat{B} = B_{g,\text{peak}}$ , the electric loading  $\bar{A} = A_{\text{rms}}$ , and the fundamental winding factor into a single figure-of-merit:

$$S_i = \frac{\pi^2}{\sqrt{2}} k_{w,1} \bar{A} \hat{B} D^2 L_e n_{\text{syn}} = C D^2 L_e n_{\text{syn}}, \quad C = \frac{\pi^2}{\sqrt{2}} k_{w,1} \bar{A} \hat{B} \quad (68)$$

The constant  $C$  depends only on the materials and the cooling capability, not on the volume; machines of different sizes can be compared at a glance through  $C$  alone.

For Design 1 with  $\hat{B} = 0.989 \text{ T}$ ,  $\bar{A} = 11.78 \text{ kA/m}$  and  $k_{w,1} = 0.9566$ :

$$C = \frac{\pi^2}{\sqrt{2}} \times 0.9566 \times 11,780 \times 0.989 \approx \mathbf{77.8 \text{ kW} \cdot \text{s/m}^3} \quad (69)$$

As a consistency check on the torque derived above,  $D = 2R_{si} = 62 \text{ mm}$  and  $n_{\text{syn}} = 1500/60 = 25 \text{ s}^{-1}$ :

$$P_{\text{out}} \approx C D^2 L_e n_{\text{syn}} = 77,800 \times (0.062)^2 \times 0.102 \times 25 \approx 762 \text{ W}, \quad (70)$$

in agreement with the shear-stress-based result of the previous subsection. The value of  $C \approx 78 \text{ kW} \cdot \text{s/m}^3$  is typical of a small, naturally-cooled industrial PMSM (Pyrhönen quotes 70–150  $\text{kW} \cdot \text{s/m}^3$  for this class of machines).

### 3.13 Number of Turns Per Phase

The number of series turns per phase is sized so that the back-EMF at rated speed equals the available phase voltage from the inverter. Assuming an SVPWM-driven three-phase voltage-source inverter on a DC bus  $V_{dc} = 48\text{ V}$ , the maximum peak phase voltage (line-to-neutral, fundamental component) in the linear modulation region is

$$\hat{V}_{ph} = \frac{V_{dc}}{\sqrt{3}} = \frac{48}{\sqrt{3}} \approx 27.7\text{ V} \quad (71)$$

The electrical angular frequency relates to the mechanical speed  $n = 1500\text{ rpm}$  through the pole-pair count  $p$ :

$$\omega_e = p \omega_m = p \cdot \frac{2\pi n}{60} = 4 \times \frac{1500 \times 2\pi}{60} = 628.3\text{ rad/s} \quad (72)$$

For a sinusoidally distributed flux linkage, the peak back-EMF is  $\hat{E}_{ph} = k_{w,1} N_{ph} \omega_e \Phi_{pole}$ ; equating this to  $\hat{V}_{ph}$  and solving for  $N_{ph}$ :

$$N_{ph} = \frac{\hat{V}_{ph}}{k_{w,1} \omega_e \Phi_{pole}} = \frac{27.7}{0.9566 \times 628.3 \times 2.18 \times 10^{-3}} \approx 21.2 \quad (73)$$

where  $k_{w,1}$  reduces the contribution of each turn to account for the spatial distribution of the conductors. The number of conductors per slot must be an integer, and is constrained by  $N_{cs} = 2N_{ph}m/N_s$  for a double-layer winding. Rounding to the nearest valid integer:

$$N_{cs} = 8 \implies N_{ph} = \frac{N_{cs} N_s}{2m} = \frac{8 \times 15}{2 \times 3} = 20 \quad (74)$$

Back-EMF at 1500 rpm:  $\hat{E}_{ph} = 0.9566 \times 628.3 \times 2.18 \times 10^{-3} \times 20 = 26.2\text{ V}$ . This leaves a margin of  $(27.7 - 26.2)/27.7 \approx 5.4\%$  to the inverter limit.

### 3.14 Phase Resistance and Copper Loss at 50°C

The Common Specs require the windings to be evaluated at  $T_{Cu} = 50^\circ\text{C}$ . Copper resistivity at this temperature is

$$\rho_{Cu}(50^\circ\text{C}) = \rho_{Cu}(20^\circ\text{C})(1 + \alpha \Delta T) = 1.72 \times 10^{-8} \times (1 + 0.00393 \times 30) \approx 1.92 \times 10^{-8} \Omega \cdot \text{m} \quad (75)$$

For the 15-slot/8-pole tooth-coil layout, each coil spans one tooth. The mean turn length is approximated as twice the axial length plus two end-turn arcs of one slot pitch each:

$$\ell_{\text{turn}} \approx 2L_e + 2\tau_s = 2 \times 102 + 2 \times 12.99 \approx 230\text{ mm} \quad (76)$$

With  $N_{ph} = 20$  series turns per phase, the total series wire length is  $L_{\text{wire}} = N_{ph} \ell_{\text{turn}} \approx 4.60\text{ m}$ . The phase resistance is

$$R_{ph} = \frac{\rho_{Cu} L_{\text{wire}}}{A_{\text{wire}}} = \frac{1.92 \times 10^{-8} \times 4.60}{3.83 \times 10^{-6}} \approx \mathbf{23\text{ m}\Omega} \quad (77)$$

The total copper loss for the three phases at the rated RMS current is

$$P_{Cu} = 3 I_{rms}^2 R_{ph} = 3 \times (19.1)^2 \times 0.023 \approx \mathbf{25\text{ W}} \quad (78)$$

Treating this as the only loss (iron and stray losses neglected at this analytical stage), the rough efficiency at rated point is

$$\eta_{\text{approx}} \approx \frac{P_{\text{out}}}{P_{\text{out}} + P_{Cu}} = \frac{762}{762 + 25} \approx \mathbf{96.8\%}. \quad (79)$$

Adding a plausible 20–30 W of iron and stray losses would bring the efficiency to the 93–95% range.

### 3.15 Summary of Analytical Results

Table 14: Analytical modelling results – Design 1

| Parameter                   | Symbol               | Value                    |
|-----------------------------|----------------------|--------------------------|
| Slot pitch                  | $\tau_s$             | 12.99 mm                 |
| Tooth width                 | $w_{tb}$             | 8.50 mm                  |
| Slot opening                | $w_s$                | 4.49 mm                  |
| Carter coefficient          | $k_c$                | 1.195                    |
| Effective air gap           | $g_{\text{eff}}$     | 5.005 mm                 |
| Effective axial length      | $L_e$                | 102 mm                   |
| Air-gap peak flux density   | $B_{g,\text{peak}}$  | 0.989 T                  |
| Flux per pole               | $\Phi_{\text{pole}}$ | 2.18 mWb                 |
| Magnetic loading            | $\bar{B}$            | 0.879 T                  |
| Stator yoke width           | $w_{sy}$             | 7.64 mm                  |
| Rotor yoke width            | $w_{ry}$             | 7.64 mm                  |
| Slot height                 | $h_{\text{slot}}$    | 11.36 mm                 |
| Electrical loading (rms)    | $A_{rms}$            | 11.78 kA/m               |
| Phase RMS current           | $I_{rms}$            | 19.1 A                   |
| Conductor area              | $A_{\text{wire}}$    | 3.83 mm <sup>2</sup>     |
| Tooth flux density (actual) | $B_{\text{tooth}}$   | 1.34 T                   |
| Yoke flux density (actual)  | $B_{\text{yoke}}$    | 1.40 T                   |
| Shear stress                | $\sigma$             | 7.87 kPa                 |
| Expected torque             | $T$                  | 4.85 N·m                 |
| Expected output power       | $P$                  | 762 W                    |
| Specific machine constant   | $C$                  | 77.8 kW·s/m <sup>3</sup> |
| Turns per phase             | $N_{ph}$             | 20                       |
| Conductors per slot         | $N_{cs}$             | 8                        |
| Phase resistance (50°C)     | $R_{ph}$             | 23 mΩ                    |
| Copper loss                 | $P_{Cu}$             | 25 W                     |
| Estimated efficiency        | $\eta$               | 96.8%                    |

## 4 Q4 – 2D FEA Modelling (ANSYS Maxwell)

### 4.1 Model Setup

#### 4.1.1 Geometry

The 2D cross-section of Design 1 is shown in Figure 7. All dimensions are taken from the analytical design (Section 3) without optimisation. The stator is modelled as a full  $360^\circ$  disk because  $\gcd(N_s, 2p) = \gcd(15, 8) = 1$ , leaving no rotational symmetry that could be exploited with master/slave boundaries. Slots are drawn as simple rectangles (no pole shoes, per the project tip allowing simplified slot geometry). Magnets are modelled as annular sectors of  $40^\circ$  mechanical arc ( $160^\circ$  electrical) at  $R_{ro,c} \leq r \leq R_{ro}$ , alternating radial magnetisation outward (N) and inward (S) around the rotor.

Design 1 -- 15-slot / 8-pole SPMSM cross-section

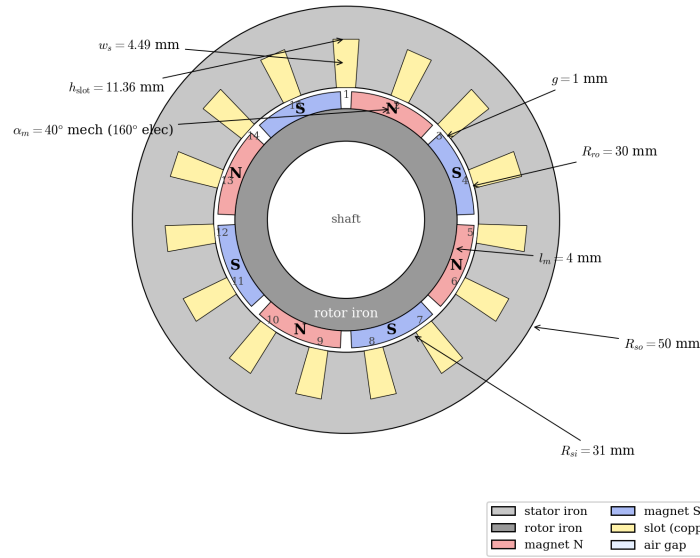


Figure 7: Cross-section of Design 1 modelled in ANSYS Maxwell 2D. 15 stator slots ( $N_s = 15$ , slot opening  $w_s = 4.49$  mm, height  $h_{slot} = 11.36$  mm), 8 surface-mount magnets ( $N_m = 8$ , arc  $40^\circ$  mech,  $l_m = 4$  mm), air gap  $g = 1$  mm, model depth  $L_e = 102$  mm.

### 4.1.2 Materials

Table 15: Material assignments in the Maxwell 2D model.

| Region                 | Library material          | Notes                                                        |
|------------------------|---------------------------|--------------------------------------------------------------|
| Stator iron            | M19_24G (Si steel)        | non-linear B–H curve                                         |
| Rotor iron             | M19_24G                   | same as stator                                               |
| Magnets ( $\times 8$ ) | Custom NdFeB              | $B_r = 1.3$ T, $\mu_r = 1.05$<br>radial mag. alternating N/S |
| Slot copper            | Copper                    | current set by Coil Terminal                                 |
| Air gap, outer air     | Vacuum                    | —                                                            |
| Shaft                  | Stainless 304 (or Vacuum) | magnetically inert                                           |

### 4.1.3 Excitations, Boundary Conditions and Mesh

**Excitations.** Three windings (A, B, C) are defined. The 15 coils are assigned according to the star-of-slots result from Section 2: Phase A on teeth 1, 3, 5, 7, 9; Phase B on teeth 6, 8, 10, 12, 14; Phase C on teeth 2, 4, 11, 13, 15. Each Coil Terminal carries  $N_{cs} = 8$  conductors.

**Boundary conditions.** A circular outer air region of radius  $1.5 R_{so} = 75$  mm is added; vector potential  $A_z = 0$  is enforced on its outer edge to contain the flux. Maxwell’s “Model Depth” setting is set to  $L_e = 102$  mm so that 2D quantities (flux, torque, EMF) come out scaled by the effective axial length automatically.

**Mesh.** The air gap is split into two thin half-gap layers to apply mesh operation to increase the calculation accuracy in the air gap. Tooth tips, magnet corners and slot openings also receive higher number of meshes.

## 4.2 No-Load Flux Density Distribution

Figure 8 shows the magnetic flux density on the full machine cross-section at no-load (zero phase current; magnets are the only source).

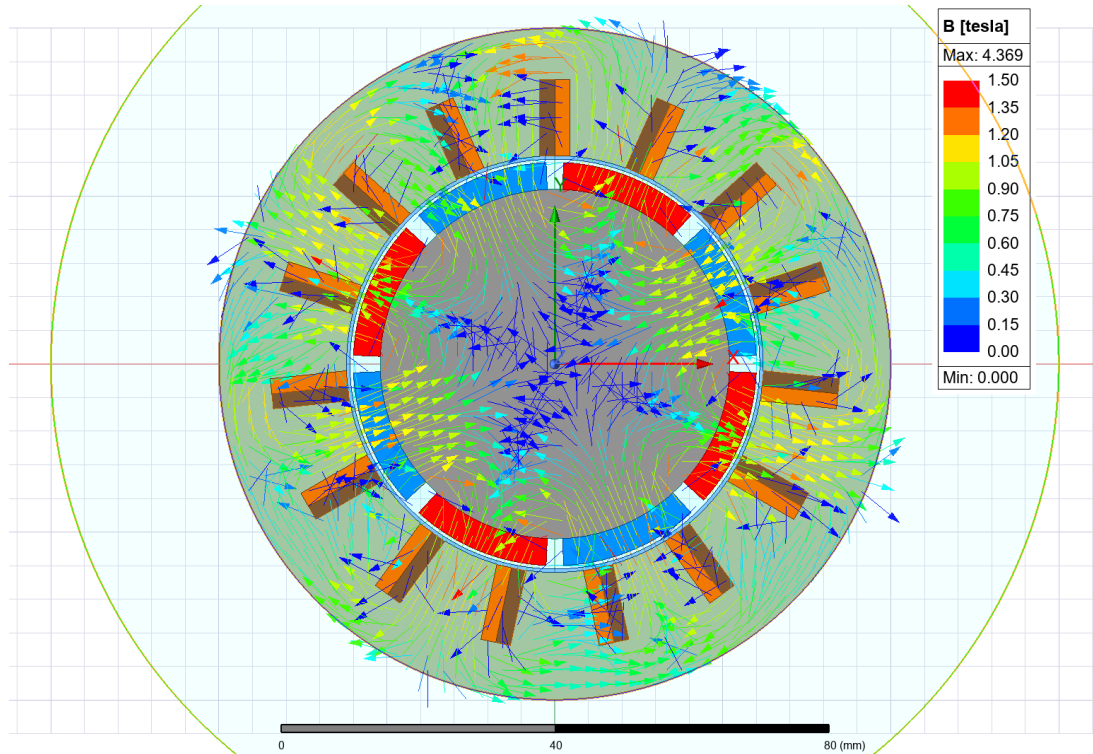


Figure 8: No-load flux density vectors over the full cross-section.

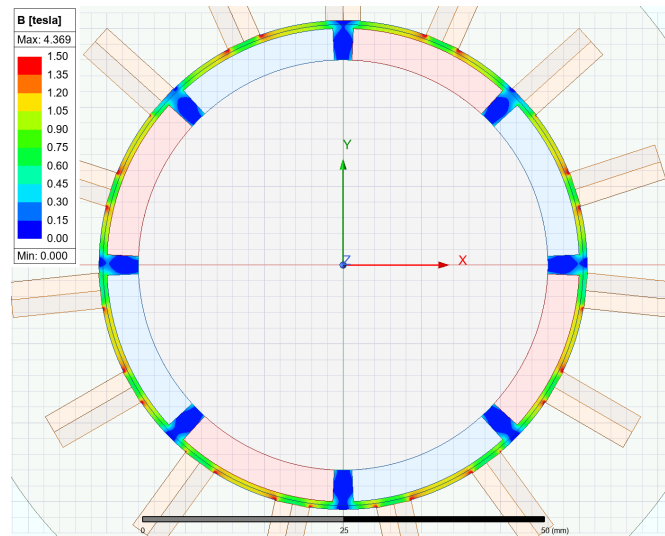
Figure 9:  $|\mathbf{B}|$  across the air-gap.

Table 16: Peak iron flux density – analytical vs. FEA (approximate values read from Figure 8).

| Region           | Analytical (Q3) | FEA (approx.) | Comment                                |
|------------------|-----------------|---------------|----------------------------------------|
| Tooth body / tip | 1.34 T          | $\sim 1.3$ T  | saturated at legend cap                |
| Stator yoke      | 1.40 T          | $\sim 1.3$ T  | below analytical (under-utilised iron) |
| Rotor yoke       | 1.40 T          | $\sim 1.3$ T  | similar                                |
| Magnet body      | —               | 1.35–1.50 T   | internal $B$ near $B_r$ , expected     |

The yokes sit slightly below the analytical target – a safe design margin – while the tooth tips push into the non-linear region locally, as expected for the geometrically narrower tooth-tip cross-section.

### 4.3 Air-Gap Flux Density

Because the 15/8 combination has no rotational symmetry ( $\gcd(15,8) = 1$ ), a single bore snapshot would mix the magnet pattern with the stator slot-permeance pattern. A cleaner approach is to fix a stator-frame probe at a tooth-tip centre and sweep the rotor over one electrical period, so each magnet passes by the same local airgap geometry. The probe sits at  $r = 30.59$  mm, lab angle  $\alpha_p = 154.44^\circ$ , and the rotor sweeps  $\theta_m = 0\text{--}90^\circ$  mechanical in  $1^\circ$  steps. At each rotor position  $B_x, B_y$  are exported and combined into  $B_r = B_x \cos \alpha_p + B_y \sin \alpha_p$ .

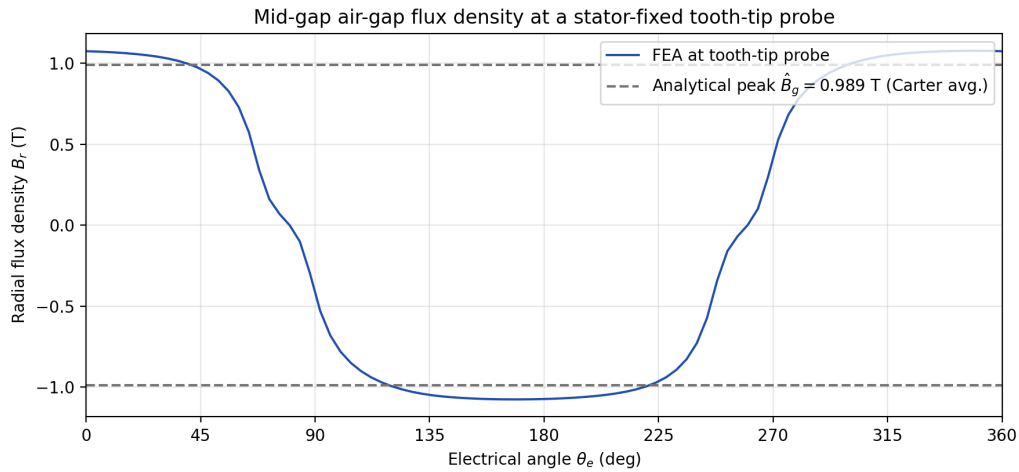


Figure 10: Mid-gap  $B_r$  at the tooth-tip probe vs. electrical angle.

The flat-top amplitude of the FEA waveform is  $\hat{B}_{g,\text{FEA}} = 1.077$  T, which exceeds the analytical peak by +8.9%. This is the expected behaviour at a tooth-tip probe: Carter's coefficient in the analytical model spatially averages the airgap permeance across one slot pitch.

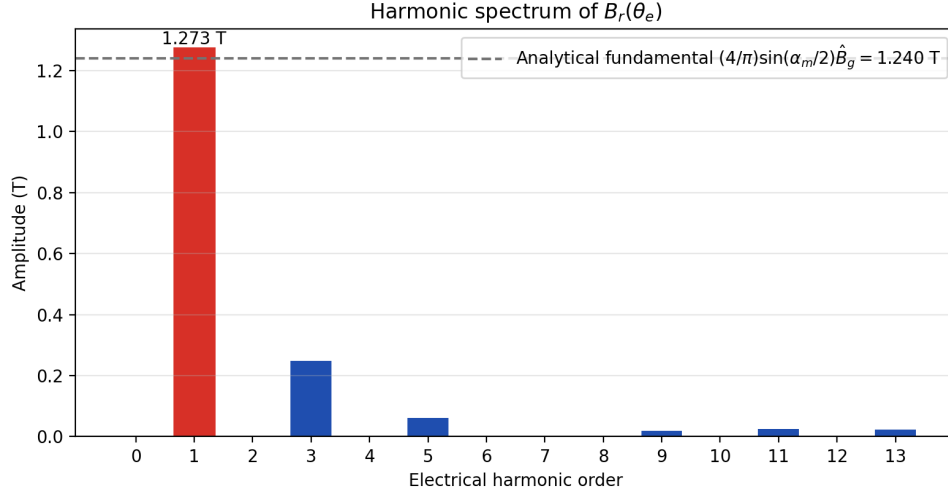
Figure 11: Harmonic spectrum of  $B_r(\theta_e)$ .

Table 17 summarises both metrics: the waveform peak (a tooth-centre value) and the fundamental Fourier component (the integrated quantity seen by the winding) – the latter agrees with the analytical  $(4/\pi)\sin(\alpha_m/2)\hat{B}_g$  within 3%.

Table 17: Air-gap flux density: analytical vs. FEA.

| Quantity                       | Analytical (Q3) | FEA (Q4) | Diff. (%) |
|--------------------------------|-----------------|----------|-----------|
| Waveform peak $\hat{B}_g$      | 0.989 T         | 1.077 T  | +8.9%     |
| Fundamental of $B_r(\theta_e)$ | 1.240 T         | 1.273 T  | +2.7%     |

#### 4.4 Leakage Flux Paths

Three distinct leakage mechanisms are visible in zoomed views of the no-load field solution: magnet-to-magnet leakage between adjacent poles, in-slot leakage threading through the slot conductor region, and tooth-tip leakage hopping across a slot opening between adjacent tooth shoes.

**Magnet-to-magnet leakage** (Figure 12.) A small portion of the magnet flux short-circuits between adjacent N and S magnets through the inter-magnet air rather than crossing into the stator iron. It is a magnet-permeance loss that contributes nothing to torque production.

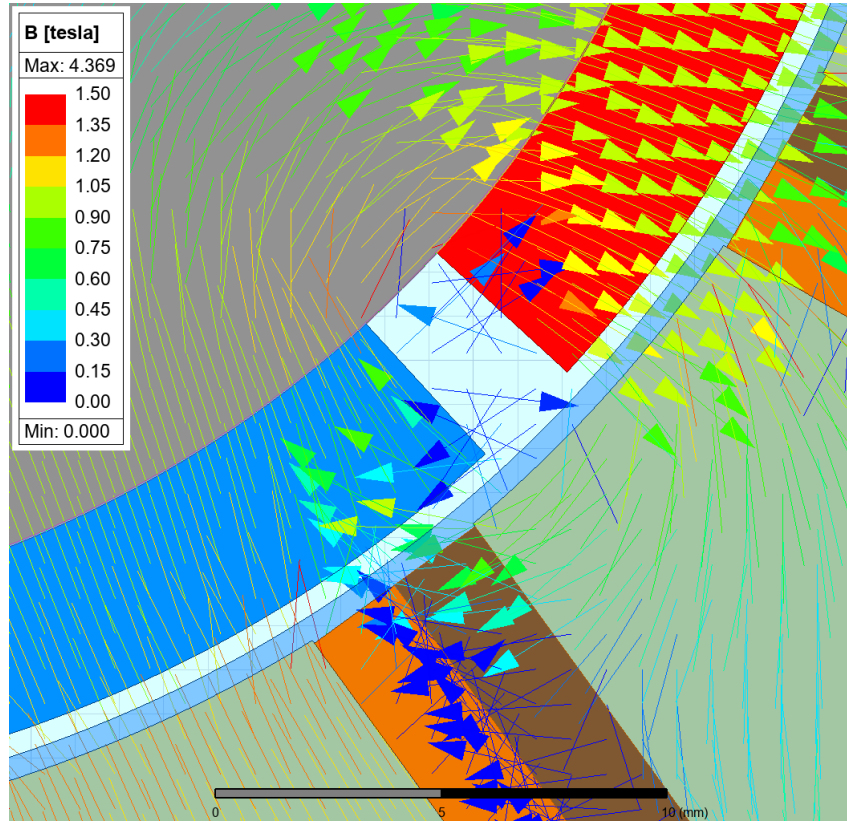


Figure 12: Magnet-to-magnet leakage.

**In-slot leakage** (Figure 13.) Flux looping inside the slot, threading from one tooth flank to the other through the conductor region. It contributes no useful torque-producing flux but stores reactive energy, appearing in lumped models as the slot-leakage inductance that raises the per-phase synchronous reactance and limits the machine's overload capability.

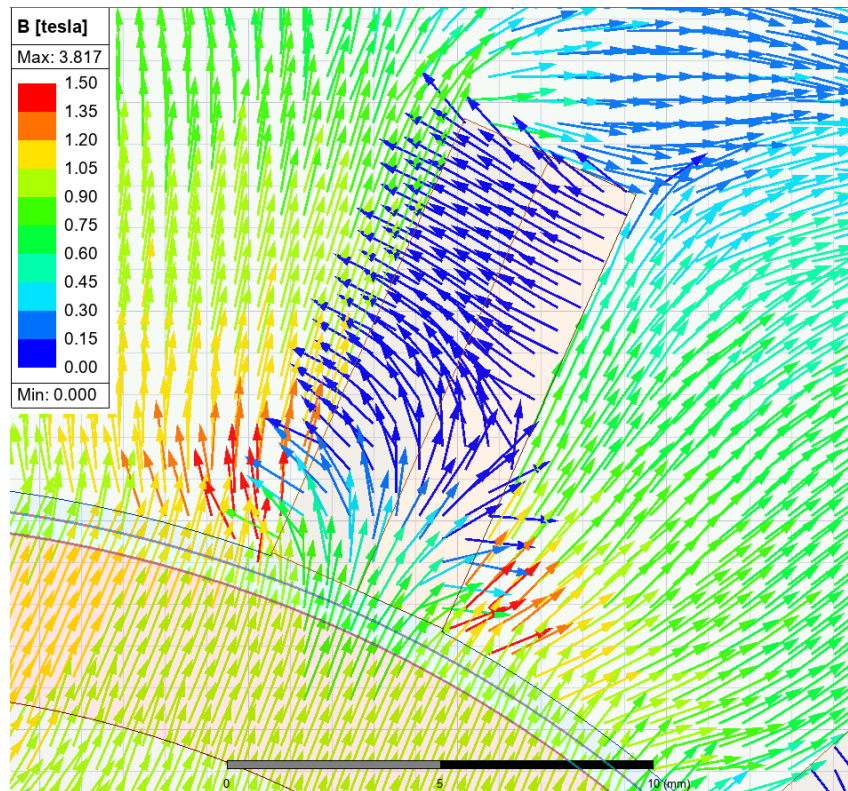


Figure 13: In-slot leakage.

**Tooth-tip leakage** (Figure 14.) A small fraction of the magnet flux hops across the slot opening between adjacent tooth-tip shoes rather than passing down the full tooth body to the yoke. This is the component Carter's coefficient parametrises in the analytical model.

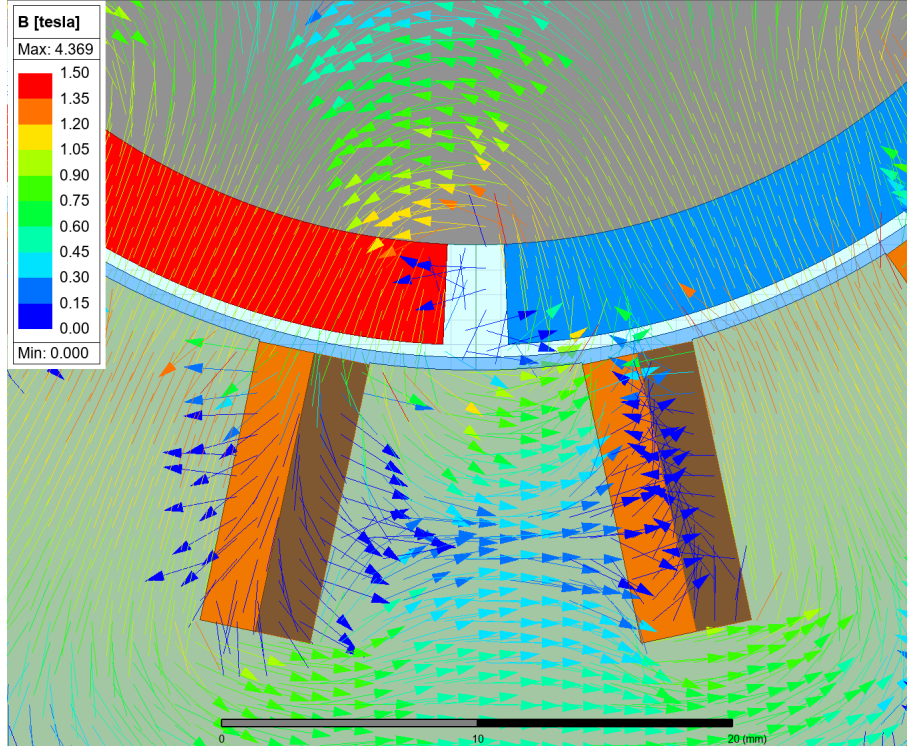


Figure 14: Tooth-tip leakage.

The magnet-to-magnet and tooth-tip paths together divert a small fraction of the magnet flux away from the useful air-gap, accounting for the  $\sim 3\%$  under-prediction of the FEA flux per pole (§4.5) relative to the analytical 1D MEC.

## 4.5 Flux per Pole

By rotational invariance of the no-load problem, the temporal  $B_r$  record from §4.3 is equivalent to a spatial snapshot along  $90^\circ$  mechanical (= 2 pole pitches) of stator arc. The flux per pole follows by spatial integration over the mid-gap arc and multiplication by the effective stack length:

$$\Phi_{pole} = \frac{1}{2} L_e R_g \int_0^{\pi/2} |B_r(\theta_m)| d\theta_m \quad (80)$$

with  $R_g = 30.5$  mm,  $L_e = 102$  mm and the  $1/2$  converting the two-pole integral to one pole.

Table 18: Flux per pole – analytical vs. FEA.

| Source                   | $\Phi_{pole}$ (mWb) | Diff. vs. analytical |
|--------------------------|---------------------|----------------------|
| Analytical (Q3)          | 2.18                | —                    |
| FEA, mid-gap integration | 2.11                | $-3.0\%$             |

The spatial integral averages high-permeance tooth regions against low-permeance slot openings – exactly the Carter average embedded in the analytical model – and the  $\sim 3\%$  residual is attributable to leakage that the 1D MEC ignores.

## 4.6 Cogging Torque (Bonus)

Cogging torque is extracted from the same parametric sweep used in §4.3: with all phase currents at zero, Maxwell evaluates the rotor torque by the Virtual Work method at each rotor position. The sweep range 0–90° mechanical contains  $90/3 = 30$  cogging cycles, since the 15/8 combination has cogging period  $\theta_{\text{cog}} = 360^\circ/\text{lcm}(15, 8) = 3^\circ$  mechanical.

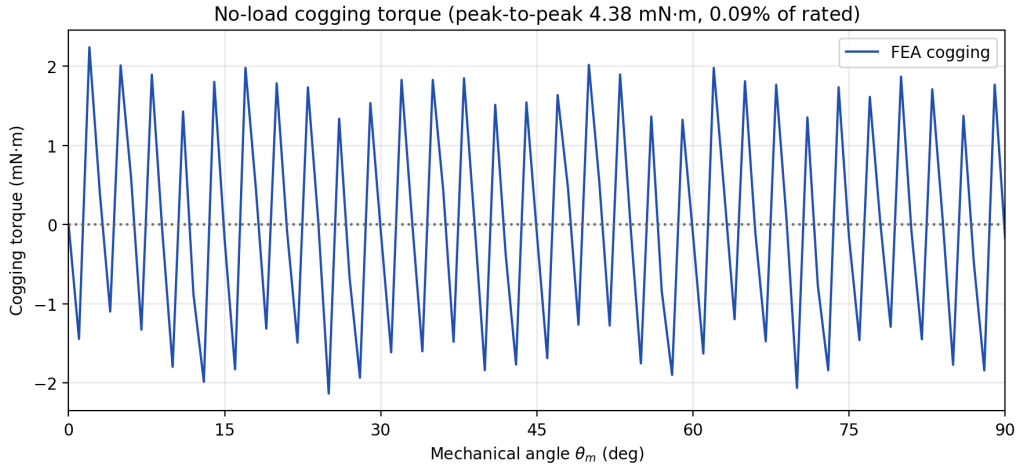


Figure 15: No-load cogging torque vs. rotor angle.

Peak-to-peak amplitude is  $T_{\text{cog,pp}} = 4.38 \text{ mN}\cdot\text{m}$  (RMS  $1.39 \text{ mN}\cdot\text{m}$ ), i.e. 0.090% of the rated torque  $4.85 \text{ N}\cdot\text{m}$  – a very low cogging level, characteristic of FSCW designs with  $\text{lcm}(15, 8) = 120$ .

## 4.7 Summary – Analytical vs. FEA

Table 19: Final comparison of analytical and FEA results for Design 1.

| Quantity                                | Analytical (Q3) | FEA (Q4)         | Diff. (%) |
|-----------------------------------------|-----------------|------------------|-----------|
| Air-gap peak $\hat{B}_g$ (waveform top) | 0.989 T         | 1.077 T          | +8.9      |
| Air-gap fundamental of $B_r$            | 1.240 T         | 1.273 T          | +2.7      |
| Flux per pole $\Phi_{\text{pole}}$      | 2.18 mWb        | 2.11 mWb         | −3.0      |
| Tooth flux density $B_{\text{tooth}}$   | 1.34 T          | ~1.3 T           | —         |
| Yoke flux density $B_{\text{yoke}}$     | 1.40 T          | ~1.3 T           | —         |
| Cogging $T_{p-p}$                       | —               | 4.4 mN·m (0.09%) | —         |

Agreement between the analytical and the 2D FEA is similar for every quantity, which is expected because Carter’s coefficient in the analytical model spatially averages over the slot pitch and so under-predicts the local tooth-tip value.

## A MATLAB Script – Analytical Calculations (Q1–Q3)

The full MATLAB script `ee568_hw2_analytical.m` is reproduced below. It computes all Q1–Q3 results, generates the winding diagrams and phasor plots in Section 2 (Figures 1–3), and prints a summary table to the console.

```

1 %% EE568 HW2 - Winding Design & Analytical Modelling
2 % Student : Sefer İDACI | ID: 2575421
3 % Design : Design 1 (ID mod 5 = 1) → Ns = 15, Nm = 8
4 % Deadline: 01 June 2026
5
6 clear; clc; close all;
7
8 %% =====
9 %% Q1 - INTEGRAL-SLOT WINDING (6-pole, 72-slot, 3-phase DL)
10 %% =====
11 fprintf('=== Q1: Integral-Slot Winding (72-slot, 6-pole, 3-phase) ===\n\n');
12
13 Ns1 = 72; p1 = 3; m = 3; % slots, pole-pairs, phases
14 ae1 = p1*360/Ns1; % elec. deg per slot = 15°
15 q1 = Ns1/(2*p1*m); % slots/pole/phase = 4
16 tau1 = Ns1/(2*p1); % pole pitch in slots = 12
17
18 fprintf(' Slot pitch (elec.) ae = %.1f°\n', ae1);
19 fprintf(' Slots/pole/phase q = %.4g\n', q1);
20 fprintf(' Pole pitch tau = %d slots\n\n', tau1);
21
22 configs = [tau1, 11]; % full-pitch, 11/12 short-pitch
23 cnames = {'Full-pitched (y=12)', '11/12 Short-pitched (y=11)'};
24 harmonics = [1, 3, 5];
25
26 for ci = 1:2
27     y = configs(ci);
28     fprintf('--- %s ---\n', cnames{ci});
29     fprintf(' %-22s %8s %8s %8s\n', 'Harmonic', 'kd', 'kp', 'kw');
30     fprintf(' %s\n', repmat('-',1,50));
31     for n = harmonics
32         kd = sin(n*q1*ae1/2 * pi/180) / (q1 * sin(n*ae1/2 * pi/180));
33         kp = sin(n*y*pi / (2*tau1));
34         kw = abs(kd) * abs(kp);
35         fprintf(' %-22s %8.4f %8.4f %8.4f\n', sprintf('n = %d',n), abs(kd), abs(kp), kw);
36     end
37     fprintf('\n');
38 end
39
40 % ---- Q1 winding diagram (one pole-pair = 24 slots, slots 1..24) ----
41 q1_slots = 2*p1*m*q1; % slots per pole-pair = 24 (one period)
42 slots_pp = 1 : q1_slots;
43
44 % Assign upper layer: sectors of 60° each, A+(0-60) C-(60-120) B+(120-180)
45 % A-(180-240) C+(240-300) B-(300-360)
46 ang1 = mod((slots_pp-1)*ae1, 360);
47 [phase1_up, dir1_up] = assign_phase(ang1); % helper at end of file
48
49 % Lower layer is shifted by (y) slots (coil return path)
50 ang1_lo = mod((slots_pp-1+configs(1))*ae1, 360); % full-pitch lower
51 [ph1_lo_fp, di1_lo_fp] = assign_phase(ang1_lo);
52 ang1_lo_sp = mod((slots_pp-1+configs(2))*ae1, 360); % short-pitch lower
53 [ph1_lo_sp, di1_lo_sp] = assign_phase(ang1_lo_sp);
54
55 figure('Name','Q1 Winding Layout - One Pole-Pair','Position',[50 50 1100 500]);
56 for ci = 1:2
57     subplot(1,2,ci);
58     if ci==1; ph_lo=ph1_lo_fp; di_lo=di1_lo_fp; else; ph_lo=ph1_lo_sp; di_lo=di1_lo_sp; end
59     draw_winding_layout(slots_pp, phase1_up, dir1_up, ph_lo, di_lo, cnames{ci});
60 end
61 saveas(gcf, 'q1_winding_layout.png');
62
63 %% =====
64 %% Q2 - FRACTIONAL-SLOT WINDING (15-slot, 8-pole, 3-phase)
65 %% =====

```

```

66 fprintf('=== Q2: Fractional-Slot Winding (15-slot, 8-pole, 3-phase) ===\n\n');
67
68 Ns = 15; Nm = 8; p = Nm/2; m = 3;
69 as = Nm*180/Ns; % electrical degrees per slot = 96°
70 q = Ns/(Nm*m); % slots per pole per phase = 0.625
71 sa = mod((0:Ns-1)*as, 360); % slot phasor angles [°], slots indexed 1..Ns
72
73 fprintf(' Electrical angle per slot as = %.2f°\n', as);
74 fprintf(' Slots per pole per phase q = %.4f\n\n', q);
75
76 % Slot angle table
77 fprintf(' Slot | Angle (°)\n');
78 fprintf(' -----\n');
79 for k = 1:Ns
80     fprintf(' %2d | %6.1f\n', k, sa(k));
81 end
82
83 % =====
84 % USER INPUT -- edit these 6 arrays to change the winding assignment
85 %
86 % Each array lists 5 slot numbers for one phase + one layer.
87 % Layer 1: first n_pos_L1 entries = positive slots (current IN, +)
88 %           remaining entries = negative slots (current OUT, -)
89 % Layer 2: first n_pos_L2 entries = positive slots
90 %           remaining entries = negative slots
91 %
92 % Slot angles: sa(k) = (k-1)*96° mod 360°
93 % k=1: 0°, k=2: 96°, k=3: 192°, k=4: 288°, k=5: 24°, k=6: 120°,
94 % k=7: 216°, k=8: 312°, k=9: 48°, k=10: 144°, k=11: 240°,
95 % k=12: 336°, k=13: 72°, k=14: 168°, k=15: 264°
96 % =====
97 n_pos_L1 = 3; % Layer 1: first 3 entries are positive (+), last 2 are negative (-)
98 n_pos_L2 = 2; % Layer 2: first 2 entries are positive (+), last 3 are negative (-)
99
100 % Winding (kw=0.9566)
101 % [ + + + - - ]
102 A_layer1 = [ 1, 5, 9, 3, 7 ];
103 B_layer1 = [ 6, 10, 14, 8, 12 ];
104 C_layer1 = [ 4, 11, 15, 2, 13 ];
105
106 % [ + + - - - ]
107 A_layer2 = [ 4, 8, 2, 6, 10 ];
108 B_layer2 = [ 9, 13, 7, 11, 15 ];
109 C_layer2 = [ 3, 14, 1, 5, 12 ];
110
111
112
113 % =====
114 % Winding factor -- phasor summation for n = 1, 3, 5
115 % =====
116 phase_labels = {'A', 'B', 'C'};
117 L1_all = {A_layer1; B_layer1; C_layer1};
118 L2_all = {A_layer2; B_layer2; C_layer2};
119
120 kw_L1 = zeros(3,3); % rows = phase (A,B,C), cols = harmonic (n=1,3,5)
121
122 for ph = 1:3
123     L1 = L1_all{ph};
124     pos_L1 = L1(1:n_pos_L1); % positive slots, layer 1
125     neg_L1 = L1(n_pos_L1+1:end); % negative slots, layer 1
126
127     for hi = 1:3
128         n = harmonics(hi);
129         % positive slots: phasor at n*angle
130         % negative slots: phasor at n*angle + 180° (current reversal)
131         ph_L1 = [n*sa(pos_L1), n*sa(neg_L1)+180];
132         kw_L1(ph,hi) = abs(sum(exp(1j * ph_L1 * pi/180))) / 5;
133     end
134 end
135
136 kw_q2 = kw_L1(1,:); % Phase A winding factors -- used in Q3
137
138 fprintf('\n Winding factors (concentrated-coil convention):\n');

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139 fprintf(' Phase |      n=1      n=3      n=5\n');
140 fprintf(' %s\n', repmat('-',1,35));
141 for ph = 1:3
142     fprintf('      %s |      %.4f      %.4f      %.4f\n', phase_labels{ph}, kw_L1(ph,:));
143 end
144
145 % =====
146 % Build per-slot phase/direction from Layer 1 (used by layout plot)
147 % ph2 values: 1=A+, 2=A-, 3=B+, 4=B-, 5=C+, 6=C-
148 % =====
149 ph2 = zeros(1,Ns);
150 dir2 = zeros(1,Ns);
151 for ph = 1:3
152     L1 = L1_all{ph};
153     ph2(L1(1:n_pos_L1)) = 2*ph-1; dir2(L1(1:n_pos_L1)) = +1;
154     ph2(L1(n_pos_L1+1:end)) = 2*ph; dir2(L1(n_pos_L1+1:end)) = -1;
155 end
156
157 % Build per-slot phase/direction from Layer 2 (for Layer 2 star plot)
158 ph2_L2 = zeros(1,Ns);
159 dir2_L2 = zeros(1,Ns);
160 for ph = 1:3
161     L2 = L2_all{ph};
162     ph2_L2(L2(1:n_pos_L2)) = 2*ph-1; dir2_L2(L2(1:n_pos_L2)) = +1;
163     ph2_L2(L2(n_pos_L2+1:end)) = 2*ph; dir2_L2(L2(n_pos_L2+1:end)) = -1;
164 end
165
166 % =====
167 % Shared plot settings (change once, applies to all figures)
168 % =====
169 cA = [0.85 0.15 0.15]; % Phase A colour
170 cB = [0.15 0.45 0.85]; % Phase B colour
171 cC = [0.15 0.70 0.20]; % Phase C colour
172 cArrow = 'b'; % phasor arrow colour
173 cResultant = 'r'; % resultant arrow colour
174 arrowWidth = 1.2; % individual phasor line width
175 resultWidth = 2.5; % resultant arrow line width
176 theta_circle = linspace(0, 2*pi, 200); % unit circle
177
178 % =====
179 % Figure 1 -- Star of slots (each slot coloured by phase)
180 % =====
181 phase_colors = {cA, cA, cB, cB, cC, cC}; % index 1..6 maps to ph2
182
183 figure('Name','Q2 Star of Slots','Position',[50 600 560 480]);
184 hold on;
185 for k = 1:Ns
186     ang = sa(k) * pi/180;
187     c = phase_colors{ph2(k)};
188     quiver(0,0, cos(ang), sin(ang), 0, ...
189         'Color',c, 'LineWidth',1.5, 'MaxHeadSize',0.3);
190     text(1.15*cos(ang), 1.15*sin(ang), num2str(k), ...
191         'FontSize',8, 'HorizontalAlignment','center', 'Color',c);
192 end
193 plot(cos(theta_circle), sin(theta_circle), 'k--', 'LineWidth',0.5);
194 axis equal; xlim([-1.4 1.4]); ylim([-1.4 1.4]); grid on;
195 xlabel('Real'); ylabel('Imaginary');
196 title('Star of Slots - 15-slot / 8-pole');
197 legend([patch(NaN,NaN,cA), patch(NaN,NaN,cB), patch(NaN,NaN,cC)], ...
198     'Phase A','Phase B','Phase C', 'Location','southeast');
199 saveas(gcf, 'q2_star_of_slots.png');
200
201 % =====
202 % Figure 2 -- Phase A phasors (n = 1, 3, 5)
203 % =====
204 pos_A = A_layer1(1:n_pos_L1);
205 neg_A = A_layer1(n_pos_L1+1:end);
206
207 figure('Name','Q2 Phase A Phasors','Position',[630 600 900 300]);
208 for hi = 1:3
209     n = harmonics(hi);
210     all_ph = [n*sa(pos_A), n*sa(neg_A)+180];
211     psum = sum(exp(1j * all_ph * pi/180));

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```

212     subplot(1,3,hi); hold on;
213     for ii = 1:length(all_ph)
214         a = all_ph(ii)*pi/180;
215         quiver(0,0, cos(a),sin(a), 0, cArrow, 'LineWidth',arrowWidth, 'MaxHeadSize',0.4);
216     end
217     quiver(0,0, real(psum)/5, imag(psum)/5, 0, ...
218         cResultant, 'LineWidth',resultWidth, 'MaxHeadSize',0.3);
219     plot(cos(theta_circle), sin(theta_circle), 'k--', 'LineWidth',0.5);
220     axis equal; xlim([-1.4 1.4]); ylim([-1.4 1.4]); grid on;
221     title(sprintf('Phase A, n=%d kw = %.4f', n, kw_L1(1,hi)));
222     xlabel('Real'); ylabel('Imaginary');
223 end
224 saveas(gcf, 'q2_phase_A_phasors.png');
225
226 % =====
227 % Figure 3 -- Stator cross-section with teeth and double-layer slots
228 % =====
229 figure('Name','Q2 Stator Winding Layout','Position',[50 100 800 800]);
230 draw_stator_winding(Ns, ph2, dir2, ph2_L2, dir2_L2, ...
231     'Design - 15-slot/8-pole, Double-Layer FSCW');
232 saveas(gcf, 'q2_winding_layout.png');
233
234 %% =====
235 %% Q3 - ANALYTICAL MODELLING
236 %% =====
237 fprintf('\n=== Q3: Analytical Modelling ===\n\n');
238
239 %--- Common specs -----
240 Rso = 50e-3; % stator outer radius [m]
241 Rro = 0.6*Rso; % rotor outer radius [m] = 30 mm
242 g = 1e-3; % air-gap length [m]
243 lm = 4e-3; % magnet radial thickness [m]
244 Br = 1.3; % remanence [T]
245 mur = 1.05; % magnet relative permeability
246 am = 160/180; % magnet arc ratio (160° elec.)
247 Ls = 100e-3; % stack length [m]
248 J = 5e6; % current density (rms) [A/m²]
249 kwc = 0.60; % conductor fill factor
250 Bt = 1.4; % target peak flux density in iron [T]
251 Vdc = 48; % DC bus voltage [V]
252 n_rpm = 1500; % rated speed [rpm]
253 kw1 = kw_q2(1); % winding factor (fundamental)
254
255 %--- Derived radii -----
256 Rsi = Rro + g; % stator inner radius [m] = 31 mm
257 Rro_core = Rro - lm; % rotor core outer radius [m] = 26 mm
258
259 fprintf('Machine Geometry\n');
260 fprintf(' Rso (stator outer) = %.1f mm\n', Rso*1e3);
261 fprintf(' Rsi (stator inner) = %.1f mm\n', Rsi*1e3);
262 fprintf(' Rro (rotor outer) = %.1f mm\n', Rro*1e3);
263 fprintf(' Rro_core (core outer) = %.1f mm\n', Rro_core*1e3);
264 fprintf(' Air gap g = %.1f mm\n', g*1e3);
265 fprintf(' Magnet thickness lm = %.1f mm\n', lm*1e3);
266
267 %--- Slot pitch at stator bore -----
268 tau_s = 2*pi*Rsi / Ns; % slot pitch [m]
269 fprintf('Stator Slot Geometry\n');
270 fprintf(' Slot pitch at bore  $\tau_s$  = %.4f mm\n', tau_s*1e3);
271
272 %--- First-pass air-gap flux density (no Carter) -----
273 Bg0 = Br*lm / (lm + mur*g);
274 fprintf('\n Bg_peak (no Carter) = %.4f T\n', Bg0);
275
276 %--- Tooth width for Bt = 1.4 T -----
277 Bg_avg0 = Bg0 * am;
278 wtb = Bg_avg0 * tau_s / Bt;
279 ws = tau_s - wtb;
280 fprintf(' Tooth body width wtb = %.4f mm\n', wtb*1e3);
281 fprintf(' Slot opening width ws = %.4f mm\n', ws*1e3);
282
283 %--- Carter's coefficient (exact formula) -----

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285 x = ws/(2*g);
286 gam = (4/pi) * (x*atan(x) - log(sqrt(1+x^2))); % length [m] when multiplied by g
287 kc = tau_s / (tau_s - gam*g);
288 fprintf('\nCarter's Coefficient\n');
289 fprintf(' ws / g = %.3f\n', ws/g);
290 fprintf(' γ (slot fringe) = %.4f mm\n', gam*g*1e3);
291 fprintf(' Carter coefficient kc = %.4f\n', kc);
292
293 %--- Effective air gap & corrected Bg -----
294 g_eff = kc*g + lm/mur;
295 Bg = Br*lm / (lm + mur*kc*g);
296 fprintf(' g_eff = kc*g + lm/μr = %.4f mm\n', g_eff*1e3);
297 fprintf(' Bg_peak (corrected) = %.4f T\n', Bg);
298
299 %--- Effective axial length (fringing correction) -----
300 Le = Ls + 2*g;
301 fprintf('\nEffective Axial Length\n');
302 fprintf(' Ls (physical) = %.1f mm\n', Ls*1e3);
303 fprintf(' Le = Ls + 2g = %.2f mm\n', Le*1e3);
304
305 %--- Pole pitch & flux per pole -----
306 tau_p = pi*Rsi / p; % pole pitch at stator bore [m]
307 Phi = Bg * am * tau_p * Le; % flux per pole [Wb]
308 fprintf('\nFlux Per Pole\n');
309 fprintf(' Pole pitch τp = %.4f mm\n', tau_p*1e3);
310 fprintf(' Flux per pole Φ = %.6f Wb (%.4f mWb)\n', Phi, Phi*1e3);
311
312 %--- Stator and rotor yoke widths -----
313 wsy = Phi / (2 * Bt * Le);
314 wry = Phi / (2 * Bt * Le);
315 fprintf('\nIron Yoke Widths\n');
316 fprintf(' Stator yoke wsy = %.4f mm\n', wsy*1e3);
317 fprintf(' Rotor yoke wry = %.4f mm\n', wry*1e3);
318 fprintf(' Rotor shaft radius = %.4f mm\n', (Rro_core - wry)*1e3);
319
320 %--- Slot dimensions & electrical loading -----
321 h_total = Rso - Rsi;
322 h_slot = h_total - wsy;
323 A_slot = ws * h_slot; % slot area (simplified rectangular) [m²]
324 A_rms = J * kwc * A_slot / tau_s; % linear current density rms [A/m]
325 A_peak = A_rms * sqrt(2);
326
327 fprintf('\nElectrical Loading\n');
328 fprintf(' Stator radial height = %.4f mm\n', h_total*1e3);
329 fprintf(' Slot height h_slot = %.4f mm\n', h_slot*1e3);
330 fprintf(' Slot area (rect.) = %.4f mm²\n', A_slot*1e6);
331 fprintf(' A_rms (linear) = %.2f kA/m\n', A_rms/1e3);
332
333 %--- Magnetic loading (average Bg over rotor surface) -----
334 Bav = Bg * am;
335 fprintf('\nMagnetic Loading\n');
336 fprintf(' Bav = Bg·am = %.4f T\n', Bav);
337
338 %--- Specific machine constant & shear stress -----
339 % Average electromagnetic shear stress: σ = kw·Bav·A_rms (N/m²)
340 % More precisely: σ = (1/2)·kw·Bg_peak·A_peak
341 sigma = 0.5 * kw1 * Bg * A_peak;
342 fprintf('\nShear Stress\n');
343 fprintf(' σ = ½·kw·Bg·A_peak = %.2f kPa\n', sigma/1e3);
344
345 %--- Expected torque & power -----
346 T_exp = sigma * 2*pi * Rsi^2 * Le;
347 omega = n_rpm * 2*pi / 60;
348 P_exp = T_exp * omega;
349 fprintf('\nPerformance Estimates\n');
350 fprintf(' Expected torque = %.3f N·m\n', T_exp);
351 fprintf(' Rated speed ω = %.2f rad/s\n', omega);
352 fprintf(' Expected power = %.1f W\n', P_exp);
353
354 %--- Number of turns per phase (for Vdc=48V, n=1500rpm) -----
355 omega_e = p * omega;
356 Vph_peak = Vdc / sqrt(3); % peak phase voltage (star, full modulation)
357 Nph_float = Vph_peak / (kw1 * omega_e * Phi);

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```

358 Ncs_float = 2*Nph_float*m / Ns; % conductors per slot (must be integer)
359 % Round to nearest integer conductors/slot
360 Ncs = round(Ncs_float);
361 Nph = Ncs * Ns / (2*m);
362 Vph_actual = kw1 * Nph * omega_e * Phi;
363
364 fprintf('\nWinding Turns (for Vdc = %.0f V, n = %.0f rpm)\n', Vdc, n_rpm);
365 fprintf('  we (elec. freq.)      = %.2f rad/s\n', omega_e);
366 fprintf('  Vph_peak (star)         = %.2f V\n', Vph_peak);
367 fprintf('  Nph (ideal)               = %.2f turns/phase\n', Nph_float);
368 fprintf('  Ncs (conductors/slot)     = %d (rounded to integer)\n', Ncs);
369 fprintf('  Nph (actual)              = %d turns/phase\n', Nph);
370 fprintf('  Back-EMF (peak)          = %.2f V\n', Vph_actual);
371
372 %--- Summary table for report -----
373 fprintf('\n=== Summary Table ===\n');
374 fprintf('  %-35s %12s\n', 'Parameter', 'Value');
375 fprintf('  %s\n', repmat('-',1,50));
376 params = {
377     'Stator inner radius Rsi',      sprintf('%.1f mm', Rsi*1e3);
378     'Effective air gap g_eff',      sprintf('%.3f mm', g_eff*1e3);
379     'Carter coefficient kc',        sprintf('%.4f', kc);
380     'Effective axial length Le',    sprintf('%.1f mm', Le*1e3);
381     'Air-gap peak flux density Bg', sprintf('%.4f T', Bg);
382     'Flux per pole Φ',              sprintf('%.4f mWb', Phi*1e3);
383     'Average flux density Bav',     sprintf('%.4f T', Bav);
384     'Electrical loading A_rms',     sprintf('%.2f kA/m', A_rms/1e3);
385     'Shear stress σ',               sprintf('%.2f kPa', sigma/1e3);
386     'Expected torque',              sprintf('%.3f N-m', T_exp);
387     'Expected power',               sprintf('%.1f W', P_exp);
388     'Turns per phase Nph',          sprintf('%d', Nph);
389 };
390 for i = 1:size(params,1)
391     fprintf('  %-35s %12s\n', params{i,1}, params{i,2});
392 end
393
394 fprintf('\nDone. Figures saved to HW2 folder.\n');
395
396 %% =====
397 %% LOCAL HELPER FUNCTIONS
398 %% =====
399
400 function [ph, d] = assign_phase(angles)
401 % Returns phase index (1=A+, 2=A-, 3=B+, 4=B-, 5=C+, 6=C-)
402 % and direction (+1 or -1) for each angle in [0,360).
403 n = length(angles);
404 ph = zeros(1,n);
405 d = zeros(1,n);
406 for k = 1:n
407     a = mod(angles(k), 360);
408     if a < 60, ph(k)=1; d(k)= 1; % A+
409     elseif a < 120, ph(k)=6; d(k)=-1; % C- (ph=6 for C-, reuse index)
410     elseif a < 180, ph(k)=3; d(k)= 1; % B+
411     elseif a < 240, ph(k)=2; d(k)=-1; % A-
412     elseif a < 300, ph(k)=5; d(k)= 1; % C+
413     else, ph(k)=4; d(k)=-1; % B-
414     end
415 end
416 % Remap to simple phase 1=A, 2=A, 3=B, 4=B, 5=C, 6=C
417 % (direction stored separately in d)
418 end
419
420 function draw_winding_layout(slots, ph_up, dir_up, ph_lo, dir_lo, ttl)
421 % Simple bar chart layout for one pole-pair of integral-slot winding.
422 cmap = [0.85 0.15 0.15; % A+ red
423         0.15 0.45 0.85; % C- blue
424         0.15 0.70 0.20; % B+ green
425         0.85 0.15 0.15; % A- red
426         0.15 0.70 0.20; % C+ green
427         0.15 0.45 0.85]; % B- blue
428 Ns = length(slots);
429 hold on;
430 for k = 1:Ns

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431     % upper layer (y = 0.5 .. 1)
432     fill([k-0.9 k-0.1 k-0.1 k-0.9], [0.5 0.5 1 1], cmap(ph_up(k,:),:), ...
433         'EdgeColor','k','LineWidth',0.5);
434     % lower layer (y = 0 .. 0.5)
435     fill([k-0.9 k-0.1 k-0.1 k-0.9], [0 0 0.5 0.5], cmap(ph_lo(k,:),:), ...
436         'EdgeColor','k','LineWidth',0.5,'FaceAlpha',0.5);
437     % direction arrows
438     if dir_up(k) > 0; sym = '\bullet'; else; sym = '\times'; end
439     text(k-0.5, 0.75, sym, 'HorizontalAlignment','center','FontSize',7);
440     if dir_lo(k) > 0; sym = '\bullet'; else; sym = '\times'; end
441     text(k-0.5, 0.25, sym, 'HorizontalAlignment','center','FontSize',7);
442 end
443 xlim([0 Ns+1]); ylim([-0.2 1.3]); axis off;
444 title(ttl, 'FontSize', 9);
445 legend([patch(NaN,NaN,[0.85 0.15 0.15]), ...
446         patch(NaN,NaN,[0.15 0.45 0.85]), ...
447         patch(NaN,NaN,[0.15 0.70 0.20])], ...
448         'Phase A','Phase B','Phase C','Location','south','FontSize',7);
449 end
450
451 function draw_winding_layout_radial(Ns, ph2, dir2, ttl)
452 % Circular slot layout for fractional-slot winding.
453 cA = [0.85 0.15 0.15];
454 cB = [0.15 0.45 0.85];
455 cC = [0.15 0.70 0.20];
456 cmap_full = {cA, cA, cB, cB, cC, cC}; % index 1..6
457
458 hold on; axis equal; axis off;
459 R = 1.0; w = 0.25;
460 angles_slot = linspace(90, 90-360, Ns+1) * pi/180;
461 for k = 1:Ns
462     th1 = angles_slot(k);
463     th2 = angles_slot(k+1);
464     th = linspace(th1, th2, 20);
465     % outer arc polygon
466     xo = R*cos(th); yo = R*sin(th);
467     xi = (R-w)*cos(fliplr(th)); yi = (R-w)*sin(fliplr(th));
468     c = cmap_full{ph2(k)};
469     fill([xo xi], [yo yi], c, 'EdgeColor','k','LineWidth',0.6);
470     % direction symbol
471     tmid = (th1+th2)/2;
472     rm = R - w/2;
473     if dir2(k) > 0; sym = '\bullet'; else; sym = '\times'; end
474     text(rm*cos(tmid), rm*sin(tmid), sym, ...
475         'HorizontalAlignment','center','FontSize',7,'Color','w','FontWeight','bold');
476     % slot number
477     text(1.15*cos(tmid), 1.15*sin(tmid), num2str(k), ...
478         'HorizontalAlignment','center','FontSize',7);
479 end
480 title(ttl, 'FontSize', 10);
481 legend([patch(NaN,NaN,cA), patch(NaN,NaN,cB), patch(NaN,NaN,cC)], ...
482         'Phase A','Phase B','Phase C','Location','south');
483 end
484
485 function draw_stator_winding(Ns, ph_L1, dir_L1, ph_L2, dir_L2, ttl)
486 % Stator cross-section for an FSCW (single-tooth concentrated winding).
487 % Each slot holds two coil sides SIDE-BY-SIDE (tangential split):
488 % - The half nearer tooth k = go-side of coil k (Layer 1)
489 % - The half nearer tooth k-1 = return-side of coil k-1 (Layer 2)
490 % Slots ordered clockwise with slot 1 at the top.
491
492 % Phase colours
493 cA = [0.95 0.55 0.55];
494 cB = [0.55 0.70 0.95];
495 cC = [0.55 0.85 0.60];
496 phase_colors = {cA, cA, cB, cB, cC, cC}; % index 1..6
497
498 % Geometry (normalised)
499 R_yoke_out = 1.00;
500 R_yoke_in = 0.85; % yoke inner edge / tooth root
501 R_bore = 0.55; % stator bore / tooth tip
502 R_rotor = 0.50;
503 tooth_frac = 0.4; % tooth angular width / slot pitch

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504
505 slot_pitch = 2*pi / Ns;
506 tooth_half = tooth_frac * slot_pitch / 2;
507 slot_half = (1-tooth_frac) * slot_pitch / 2;
508 theta0 = pi/2;
509
510 hold on; axis equal; axis off;
511 th_full = linspace(0, 2*pi, 360);
512
513 % --- Back-iron annulus -----
514 fill([R_yoke_out*cos(th_full), fliplr(R_yoke_in*cos(th_full))], ...
515      [R_yoke_out*sin(th_full), fliplr(R_yoke_in*sin(th_full))], ...
516      [0.80 0.80 0.80], 'EdgeColor','k','LineWidth',0.8);
517
518 % --- Teeth -----
519 for k = 1:Ns
520     slot_c = theta0 - (k-1)*slot_pitch;
521     tooth_c = slot_c - slot_pitch/2; % tooth k between slot k and slot k+1
522     th_t = linspace(tooth_c - tooth_half, tooth_c + tooth_half, 20);
523     xo = R_yoke_in*cos(th_t); yo = R_yoke_in*sin(th_t);
524     xi = R_bore *cos(fliplr(th_t)); yi = R_bore *sin(fliplr(th_t));
525     fill([xo xi], [yo yi], [0.75 0.75 0.75], ...
526          'EdgeColor','k','LineWidth',0.6);
527 end
528
529 % --- Slot fillings (TANGENTIAL split: L1 right side, L2 left side)
530 phase_letter = {'A','A','B','B','C','C'};
531 for k = 1:Ns
532     slot_c = theta0 - (k-1)*slot_pitch;
533
534     % Slot 1 is at theta = pi/2 (top); slots advance CLOCKWISE so
535     % "smaller theta" is to the RIGHT of the slot (closer to tooth k),
536     % "larger theta" is to the LEFT (closer to tooth k-1).
537     th_R = linspace(slot_c - slot_half, slot_c, 14); % right side (Layer 1)
538     th_L = linspace(slot_c, slot_c + slot_half, 14); % left side (Layer 2)
539
540     % --- Right side of slot = Layer 1 (go-side of coil k) -----
541     c1 = phase_colors{ph_L1(k)};
542     xo = R_yoke_in*cos(th_R); yo = R_yoke_in*sin(th_R);
543     xi = R_bore *cos(fliplr(th_R)); yi = R_bore *sin(fliplr(th_R));
544     fill([xo xi], [yo yi], c1, 'EdgeColor','k','LineWidth',0.4);
545
546     % --- Left side of slot = Layer 2 (return of coil k-1) -----
547     c2 = phase_colors{ph_L2(k)};
548     xo = R_yoke_in*cos(th_L); yo = R_yoke_in*sin(th_L);
549     xi = R_bore *cos(fliplr(th_L)); yi = R_bore *sin(fliplr(th_L));
550     fill([xo xi], [yo yi], c2, 'EdgeColor','k','LineWidth',0.4);
551
552     % --- Radial separator line between the two coil sides -----
553     plot([R_yoke_in*cos(slot_c), R_bore*cos(slot_c)], ...
554          [R_yoke_in*sin(slot_c), R_bore*sin(slot_c)], ...
555          'k-', 'LineWidth',0.6);
556
557     % --- Direction symbols and phase labels -----
558     rm = (R_yoke_in + R_bore)/2;
559     thR = slot_c - slot_half/2; % centre of right half (Layer 1)
560     thL = slot_c + slot_half/2; % centre of left half (Layer 2)
561     if dir_L1(k) > 0, s1 = '\bullet'; else, s1 = '\times'; end
562     if dir_L2(k) > 0, s2 = '\bullet'; else, s2 = '\times'; end
563     sgn1 = '+'; if dir_L1(k) < 0, sgn1 = char(8722); end
564     sgn2 = '+'; if dir_L2(k) < 0, sgn2 = char(8722); end
565
566     text(rm*cos(thR), rm*sin(thR), ...
567          {s1, [phase_letter{ph_L1(k)} sgn1]}, ...
568          'HorizontalAlignment','center','VerticalAlignment','middle','FontSize',7);
569     text(rm*cos(thL), rm*sin(thL), ...
570          {s2, [phase_letter{ph_L2(k)} sgn2]}, ...
571          'HorizontalAlignment','center','VerticalAlignment','middle','FontSize',7);
572
573     % --- Slot number -----
574     text(1.10*cos(slot_c), 1.10*sin(slot_c), num2str(k), ...
575          'HorizontalAlignment','center','FontSize',9,'FontWeight','bold');
576 end

```

```

577
578 % --- Rotor (visual reference) -----
579 fill(R_rotor*cos(th_full), R_rotor*sin(th_full), [0.95 0.90 0.75], ...
580      'EdgeColor','k','LineWidth',0.6);
581 text(0, 0, 'Rotor', 'HorizontalAlignment','center','FontSize',8, ...
582      'Color',[0.4 0.4 0.4]);
583
584 xlim([-1.25 1.25]); ylim([-1.25 1.25]);
585 title(ttl, 'FontSize', 11);
586 legend([patch(NaN,NaN,cA), patch(NaN,NaN,cB), patch(NaN,NaN,cC)], ...
587        'Phase A','Phase B','Phase C', ...
588        'Location','southoutside','Orientation','horizontal');
589 end

```

## B Python Script – Q4 FEA Post-processing

Reads the Maxwell parametric CSV exports (B-field at the probe + cogging torque), projects  $B_r$ , integrates the flux per pole, generates Figures 10, 11 and 15, and prints the comparison numbers used in Tables 17, 18 and 19.

```

1  """Post-process the no-load parametric sweep exported from ANSYS Maxwell 2D.
2
3  The sweep rotates the rotor over one electrical period (rotor_angle = 0..90 deg
4  mech, 8-pole machine) while observing a stationary probe in the airgap.
5  At each rotor position, the parametric solve records:
6
7  * B_x, B_y at the probe point (from a Fields Report)
8  * Cogging torque on the rotor (from a Parameter)
9
10 This script projects (B_x, B_y) -> B_r using the constant probe angle in
11 the lab frame, plots the waveform, FFT, and cogging torque, and prints a
12 comparison to the analytical Q3 values.
13
14 Output figures:
15   q4_bg_theta.png    -- B_r vs electrical angle
16   q4_bg_fft.png      -- harmonic spectrum (electrical orders)
17   q4_cogging.png     -- cogging torque vs mechanical angle
18 """
19
20 from pathlib import Path
21
22 import numpy as np
23 import matplotlib.pyplot as plt
24
25 # =====
26 #                               Inputs
27 # =====
28 HW2_DIR = Path(__file__).parent
29 CSV_B   = HW2_DIR / "q4_bfield_sweep.csv"
30 CSV_T   = HW2_DIR / "q4_cogging_sweep.csv"
31
32 # Probe location in the LAB (Global) frame, mm
33 PROBE_X_MM = -27.6
34 PROBE_Y_MM = 13.2
35 ALPHA_PROBE_RAD = np.arctan2(PROBE_Y_MM, PROBE_X_MM) # lab-frame angle
36 R_PROBE_MM      = np.hypot(PROBE_X_MM, PROBE_Y_MM)
37
38 POLE_PAIRS = 4          # 8 poles
39 N_POLES    = 8
40 B_ANALYTICAL_PEAK = 0.989 # T (Q3 analytical Bg waveform peak with Carter)
41 MAGNET_ARC_ELEC_DEG = 160.0 # electrical degrees of magnet pole arc
42 # Analytical fundamental of the rectangular Bg waveform = (4/pi)*sin(alpha_m*pi/2)*Bpeak
43 B_ANALYTICAL_FUND = (4.0 / np.pi) * np.sin(np.deg2rad(MAGNET_ARC_ELEC_DEG) / 2.0) * B_ANALYTICAL_PEAK
44 T_ANALYTICAL = 4.85      # Nm (Q3 rated -- for context only; no-load avg is ~0)
45 PHI_ANALYTICAL = 2.18e-3 # Wb (Q3 analytical flux per pole)
46
47 # Machine geometry / winding (from Q3)

```

```

48 R_GAP_M = 30.5e-3          # m, mid-gap radius
49 L_E_M   = 102e-3          # m, effective stack length
50 N_PH     = 20              # series turns per phase
51 KW1_ANALYTICAL = 0.9566   # Q2 analytical fundamental winding factor
52
53 N_HARM_PLOT = 13           # how many electrical harmonics to show on the bar chart
54
55
56 # =====
57 def load_b_csv(path: Path):
58     """Return (theta_mech_deg, Bx, By) from the Maxwell parametric CSV.
59     Columns: rotor_angle[deg], stator_angle[deg], Bx_probe[T], By_probe[T]."""
60     arr = np.loadtxt(path, delimiter=",", skiprows=1)
61     theta_mech_deg = arr[:, 0]
62     bx = arr[:, 2]
63     by = arr[:, 3]
64     return theta_mech_deg, bx, by
65
66
67 def load_torque_csv(path: Path):
68     """Return (theta_mech_deg, torque_Nm). Strips 'deg' and 'NewtonMeter' suffixes."""
69     theta = []
70     tau = []
71     with open(path, "r", encoding="utf-8") as f:
72         next(f) # header
73         for line in f:
74             parts = line.strip().split(",")
75             if len(parts) < 3:
76                 continue
77             theta.append(float(parts[1].replace("deg", "")))
78             tau.append(float(parts[2].replace("NewtonMeter", "")))
79     return np.array(theta), np.array(tau)
80
81
82 def load_flux_linkage_csv(path: Path):
83     """Return (theta_mech_deg, psi_A, psi_B, psi_C) in Wb."""
84     theta = []
85     psi_a = []
86     psi_b = []
87     psi_c = []
88     with open(path, "r", encoding="utf-8") as f:
89         next(f) # header
90         for line in f:
91             parts = line.strip().split(",")
92             if len(parts) < 6:
93                 continue
94             theta.append(float(parts[1].replace("deg", "")))
95             psi_a.append(float(parts[3].replace("Wb", "")))
96             psi_b.append(float(parts[4].replace("Wb", "")))
97             psi_c.append(float(parts[5].replace("Wb", "")))
98     return (np.array(theta), np.array(psi_a),
99           np.array(psi_b), np.array(psi_c))
100
101
102 def project_radial(bx, by, alpha_rad):
103     return bx * np.cos(alpha_rad) + by * np.sin(alpha_rad)
104
105
106 def flux_per_pole(theta_mech_deg, br):
107     """Flux per pole by spatial integration of B_r along the airgap.
108
109     By the rotational symmetry of the no-load problem (the stator is fixed,
110     only the rotor rotates), the temporal record of B_r at a stationary
111     probe as the rotor sweeps through one electrical period is identical
112     to a spatial snapshot of B_r along one electrical-period arc at the
113     airgap at a fixed time. So integrating |B_r| over the 0-90 deg
114     rotor-angle range (= 2 pole pitches) and dividing by 2 gives the
115     average flux per pole.
116
117     Phi_pole = (1/2) * L_e * R_gap * Integral_{0}^{pi/2} |B_r| dtheta_m
118     """
119     theta_rad = np.deg2rad(theta_mech_deg)
120     integral_abs = np.trapezoid(np.abs(br), theta_rad) # over 0..pi/2

```

```

121     n_poles_swept = N_POLES * (theta_mech_deg[-1] - theta_mech_deg[0]) / 360.0
122     phi = R_GAP_M * L_E_M * integral_abs / n_poles_swept
123     return phi
124
125
126 def fft_electrical(theta_mech_deg, br):
127     """FFT in electrical order. Theta sweep covers one full electrical period
128     (90 deg mech = 360 deg elec for an 8-pole machine), so the discrete FFT
129     bins map directly to electrical harmonic orders k = 1, 2, 3, ..."""
130     # Drop the wrap point if rotor angle ends exactly at 90 deg
131     if np.isclose(theta_mech_deg[-1], 90.0):
132         theta_mech_deg = theta_mech_deg[:-1]
133         br = br[:-1]
134     N = len(br)
135     fft_c = np.fft.rfft(br)
136     amp = 2.0 * np.abs(fft_c) / N
137     amp[0] /= 2.0
138     return np.arange(len(amp)), amp
139
140
141 def plot_br_vs_theta(theta_mech_deg, br, out_path):
142     theta_elec = POLE_PAIRS * theta_mech_deg
143     fig, ax = plt.subplots(figsize=(9, 4.3))
144     ax.plot(theta_elec, br, lw=1.4, color="#1f4eaf", label="FEA at tooth-tip probe")
145     ax.axhline(B_ANALYTICAL_PEAK, ls="--", color="0.45",
146               label=f"Analytical peak  $\hat{B}_g = \{B\_ANALYTICAL\_PEAK\}$  T (Carter avg.)")
147     ax.axhline(-B_ANALYTICAL_PEAK, ls="--", color="0.45")
148     ax.set_xlabel(r"Electrical angle  $\theta_e$  (deg)")
149     ax.set_ylabel(r"Radial flux density  $B_r$  (T)")
150     ax.set_title("Mid-gap air-gap flux density at a stator-fixed tooth-tip probe")
151     ax.set_xlim(0, 360)
152     ax.set_xticks(np.arange(0, 361, 45))
153     ax.grid(alpha=0.3)
154     ax.legend(loc="upper right", fontsize=10)
155     fig.tight_layout()
156     fig.savefig(out_path, dpi=200, bbox_inches="tight")
157     plt.close(fig)
158     print(f"Wrote {out_path}")
159
160
161 def plot_harmonics(orders, amplitudes, out_path):
162     fig, ax = plt.subplots(figsize=(8, 4.3))
163     n_show = min(N_HARM_PLOT, len(amplitudes) - 1)
164     bars = ax.bar(orders[:n_show + 1], amplitudes[:n_show + 1],
165                  color="#1f4eaf", width=0.7)
166     if len(bars) > 1:
167         bars[1].set_color("#d73027") # electrical fundamental
168         ax.text(1, amplitudes[1], f"{amplitudes[1]:.3f} T",
169               ha="center", va="bottom", fontsize=10)
170     ax.axhline(B_ANALYTICAL_FUND, ls="--", color="0.45",
171               label=f"Analytical fundamental "
172                    f" $r\hat{f}(\frac{4}{\pi})\sin(\frac{\alpha_m}{2})\hat{B}_g = \{B\_ANALYTICAL\_FUND:.3f\}$  T)")
173     ax.set_xlabel("Electrical harmonic order")
174     ax.set_ylabel("Amplitude (T)")
175     ax.set_title(r"Harmonic spectrum of  $B_r(\theta_e)$ ")
176     ax.set_xticks(orders[:n_show + 1])
177     ax.grid(alpha=0.3, axis="y")
178     ax.legend(loc="upper right", fontsize=10)
179     fig.tight_layout()
180     fig.savefig(out_path, dpi=200, bbox_inches="tight")
181     plt.close(fig)
182     print(f"Wrote {out_path}")
183
184
185 def plot_flux_linkages(theta_mech_deg, psi_a, psi_b, psi_c, out_path):
186     """Plot the three phase flux linkages over one electrical period.
187     Each is plotted both raw (with whatever DC offset Maxwell stores) and
188     relative to its own mean -- the AC part is the physically meaningful
189     rotor-driven linkage."""
190     theta_e = POLE_PAIRS * theta_mech_deg
191
192     fig, ax = plt.subplots(figsize=(9, 4.3))
193     for name, psi, color in [

```

```

194     ("$\psi_A$", psi_a, "#1f4eaf"),
195     ("$\psi_B$", psi_b, "#d73027"),
196     ("$\psi_C$", psi_c, "#2ca25f"),
197 ]:
198     ax.plot(theta_e, (psi - psi.mean()) * 1e3, lw=1.4,
199             color=color, label=name)
200 ax.set_xlabel(r"Electrical angle $\theta_e$ (deg)")
201 ax.set_ylabel("Phase flux linkage (mWb), AC component")
202 ax.set_title("No-load magnet flux linkage per phase (DC offset removed)")
203 ax.set_xlim(0, 360)
204 ax.set_xticks(np.arange(0, 361, 45))
205 ax.axhline(0, ls=":", color="0.4")
206 ax.grid(alpha=0.3)
207 ax.legend(loc="upper right", fontsize=10, ncol=3)
208 fig.tight_layout()
209 fig.savefig(out_path, dpi=200, bbox_inches="tight")
210 plt.close(fig)
211 print(f"Wrote {out_path}")
212
213
214 def winding_factor_from_psi_and_bg(psi_fund_peak_Wb, b_fund_T):
215     """Back-calculate  $k_{w,1}$  from FEA fundamental flux linkage and fundamental  $B_g$ .
216
217     For a sinusoidal radial flux density at the airgap with fundamental
218     peak  $B_{g,1}$ , the flux through one pole pitch is:
219          $\Phi_{pole,1} = 2 * B_{g,1} * R_{gap} * L_e / p$ 
220     The peak phase flux linkage on harmonic 1 is:
221          $\psi_{A,1-peak} = N_{ph} * k_{w,1} * \Phi_{pole,1}$ 
222     Hence:
223          $k_{w,1} = \psi_{A,1-peak} / (N_{ph} * \Phi_{pole,1})$ 
224     """
225     phi_pole_1 = 2.0 * b_fund_T * R_GAP_M * L_E_M / POLE_PAIRS
226     return psi_fund_peak_Wb / (N_PH * phi_pole_1), phi_pole_1
227
228
229 def plot_cogging(theta_mech_deg, tau_Nm, out_path):
230     fig, ax = plt.subplots(figsize=(9, 4.3))
231     ax.plot(theta_mech_deg, tau_Nm * 1e3, lw=1.4, color="#1f4eaf", label="FEA cogging")
232     ax.axhline(0, ls=":", color="0.4")
233     ax.set_xlabel(r"Mechanical angle $\theta_m$ (deg)")
234     ax.set_ylabel("Cogging torque (mN$\cdot$m)")
235     pp = (tau_Nm.max() - tau_Nm.min()) * 1e3
236     ax.set_title(f"No-load cogging torque (peak-to-peak {pp:.2f} mN$\cdot$m, "
237                 f"{100*pp*1e-3/T_ANALYTICAL:.2f}% of rated)")
238     ax.set_xlim(0, 90)
239     ax.set_xticks(np.arange(0, 91, 15))
240     ax.grid(alpha=0.3)
241     ax.legend(loc="upper right", fontsize=10)
242     fig.tight_layout()
243     fig.savefig(out_path, dpi=200, bbox_inches="tight")
244     plt.close(fig)
245     print(f"Wrote {out_path}")
246
247
248 # =====
249 if __name__ == "__main__":
250     print(f"Probe at ({PROBE_X_MM}, {PROBE_Y_MM}) mm -> "
251           f"r = {R_PROBE_MM:.3f} mm, alpha = {np.degrees(ALPHA_PROBE_RAD):.3f} deg")
252     print(f"Projection coeffs: cos(alpha) = {np.cos(ALPHA_PROBE_RAD):+.4f}, "
253           f"sin(alpha) = {np.sin(ALPHA_PROBE_RAD):+.4f}")
254     print()
255
256     # ---- B field -----
257     theta_mech_deg, bx, by = load_b_csv(CSV_B)
258     br = project_radial(bx, by, ALPHA_PROBE_RAD)
259
260     print(f"|B|_peak at probe : {np.hypot(bx, by).max():.4f} T")
261     print(f"B_r peak (positive) : {br.max():+.4f} T")
262     print(f"B_r peak (negative) : {br.min():+.4f} T")
263     print()
264
265     plot_br_vs_theta(theta_mech_deg, br, HW2_DIR / "q4-bg_theta.png")
266

```

```

267 orders, amps = fft_electrical(theta_mech_deg, br)
268 plot_harmonics(orders, amps, HW2_DIR / "q4_bg_fft.png")
269
270 print(" k | amplitude (T) | comment")
271 print("-----+-----")
272 for n in range(min(N_HARM_PLOT + 1, len(amps))):
273     comment = ""
274     if n == 0:
275         comment = "DC (should be ~0)"
276     elif n == 1:
277         comment = "<- electrical fundamental"
278     elif n in (3, 5, 7):
279         comment = "magnet shape odd harmonic"
280     elif n == 15:
281         comment = "slot-pole interaction"
282     print(f"{n:2d}| {amps[n]:11.4f} | {comment}")
283
284 print()
285 fea_peak = max(br.max(), -br.min())
286 print("---- Comparison to analytical (Q3) ----")
287 print(f"Waveform peak (flat-top of square-wave shape):")
288 print(f"   FEA at tooth-tip probe : {fea_peak:.4f} T")
289 print(f"   Analytical (Carter avg.): {B_ANALYTICAL_PEAK:.4f} T")
290 print(f"   Difference                : {100*(fea_peak-B_ANALYTICAL_PEAK)/B_ANALYTICAL_PEAK:+.2f}% "
291       f"(tooth-tip reads above spatial average)")
292 print()
293 print(f"Fundamental Fourier component of Bg(theta_e):")
294 print(f"   FEA                        : {amps[1]:.4f} T")
295 print(f"   Analytical (4/pi)*sin(am/2)*Bpeak : {B_ANALYTICAL_FUND:.4f} T")
296 print(f"   Difference                  : {100*(amps[1]-B_ANALYTICAL_FUND)/B_ANALYTICAL_FUND:+.2f}%")
297 print()
298
299 # ---- Cogging torque -----
300 theta_t, tau = load_torque_csv(CSV_T)
301 plot_cogging(theta_t, tau, HW2_DIR / "q4_cogging.png")
302
303 print(f"Cogging torque, peak-to-peak : {(tau.max()-tau.min())*1e3:.3f} mN.m")
304 print(f"Cogging RMS                    : {np.std(tau)*1e3:.3f} mN.m")
305 print(f"Rated torque (analytical)      : {T_ANALYTICAL*1e3:.1f} mN.m")
306 print(f"Cogging p-p / rated            : {100*(tau.max()-tau.min())/T_ANALYTICAL:.3f} %")
307
308 # ---- Flux per pole -----
309 phi_fea = flux_per_pole(theta_mech_deg, br)
310 print()
311 print("---- Flux per pole (mid-gap spatial integration) ----")
312 print(f"FEA                : {phi_fea*1e3:.4f} mWb")
313 print(f"Analytical: {PHI_ANALYTICAL*1e3:.4f} mWb")
314 print(f"Difference: {100*(phi_fea-PHI_ANALYTICAL)/PHI_ANALYTICAL:+.2f}%")

```

## C Python Script – Q3 Diagrams

Generates the Q3 conceptual figures (4, 5, 6).

```

1  """Schematic diagrams for the Q3 analytical-modelling section.
2
3  Produces three PNGs used in the EE568 HW2 report:
4      q3_geometry_sketch.png -- developed slot/tooth/magnet geometry
5      q3_mec.png             -- 1D magnetic equivalent circuit
6      q3_flux_funnel.png     -- flux funnelling from slot pitch into tooth
7
8  Run from the HW2 folder: python q3_diagrams.py
9  """
10
11 from pathlib import Path
12
13 import matplotlib.pyplot as plt
14 from matplotlib.patches import FancyArrowPatch, FancyBboxPatch, Polygon, Rectangle
15

```

```

16 OUT_DIR = Path(__file__).parent
17 DPI = 200
18
19 plt.rcParams.update({
20     "font.family": "serif",
21     "font.size": 16,
22     "mathtext.fontset": "cm",
23     "axes.linewidth": 0.8,
24 })
25
26
27 # -----
28 # 1. Slot / tooth geometry (developed view)
29 # -----
30 def geometry_sketch():
31     fig, ax = plt.subplots(figsize=(9.0, 5.6))
32
33     # Layers (y-extents)
34     rotor_y = (-0.9, 0.0)
35     magnet_y = (0.0, 1.2)
36     airgap_y = (1.2, 1.5)
37     teeth_y = (1.5, 3.0)
38     yoke_y = (3.0, 3.6)
39
40     x_left, x_right = -4.2, 4.2
41
42     # Stator yoke (label placed on the right side, outside the body)
43     ax.add_patch(Rectangle((x_left, yoke_y[0]), x_right - x_left,
44                             yoke_y[1] - yoke_y[0], facecolor="0.75",
45                             edgecolor="black", lw=0.8))
46     ax.annotate("stator yoke", xy=(x_right, 0.5 * (yoke_y[0] + yoke_y[1])),
47                 xytext=(x_right + 0.6, 0.5 * (yoke_y[0] + yoke_y[1])),
48                 ha="left", va="center", fontsize=15,
49                 arrowprops=dict(arrowstyle="-", lw=0.6))
50
51     # Teeth + slot (linearised view over one slot pitch centred on x=0)
52     slot_half = 0.75 # half slot opening width
53     tau_half = 2.10 # half slot pitch (tooth-centre to tooth-centre)
54
55     # Left tooth (extends from x_left up to slot edge)
56     ax.add_patch(Rectangle((-3.5, teeth_y[0]), 3.5 - slot_half,
57                             teeth_y[1] - teeth_y[0],
58                             facecolor="0.88", edgecolor="black", lw=0.8))
59
60     # Slot
61     ax.add_patch(Rectangle((-slot_half, teeth_y[0]), 2 * slot_half,
62                             teeth_y[1] - teeth_y[0],
63                             facecolor="#fff2a8", edgecolor="black", lw=0.8))
64
65     # Right tooth
66     ax.add_patch(Rectangle((slot_half, teeth_y[0]), 3.5 - slot_half,
67                             teeth_y[1] - teeth_y[0],
68                             facecolor="0.88", edgecolor="black", lw=0.8))
69
70     # Tooth / slot labels (inside the bodies)
71     ax.text(-2.1, 2.25, "tooth", ha="center", va="center", fontsize=15)
72     ax.text(0.0, 2.25, "slot", ha="center", va="center", fontsize=15)
73     ax.text(2.1, 2.25, "tooth", ha="center", va="center", fontsize=15)
74
75     # Tooth-centre dashed guide lines (to anchor the slot pitch dimension)
76     for x in (-tau_half, tau_half):
77         ax.plot([x, x], [teeth_y[0], 4.55], linestyle=":",
78                 color="0.45", lw=0.7)
79
80     # Air gap
81     ax.add_patch(Rectangle((x_left, airgap_y[0]), x_right - x_left,
82                             airgap_y[1] - airgap_y[0],
83                             facecolor="#d6e7ff", edgecolor="black", lw=0.4))
84     ax.annotate("air gap $g$", xy=(x_right, 0.5 * (airgap_y[0] + airgap_y[1])),
85                 xytext=(x_right + 0.6, 0.5 * (airgap_y[0] + airgap_y[1])),
86                 ha="left", va="center", fontsize=15,
87                 arrowprops=dict(arrowstyle="-", lw=0.6))
88
89     # Magnet
90     ax.add_patch(Rectangle((x_left, magnet_y[0]), x_right - x_left,

```

```

89         magnet_y[1] - magnet_y[0],
90         facecolor="#f4c6c6", edgecolor="black", lw=0.8))
91     ax.text(0, 0.5 * (magnet_y[0] + magnet_y[1]),
92             r"Magnet ($B_r$, $l_m$, $\mu_r$)",
93             ha="center", va="center")
94
95     # Rotor core
96     ax.add_patch(Rectangle((x_left, rotor_y[0]), x_right - x_left,
97                             rotor_y[1] - rotor_y[0],
98                             facecolor="0.55", edgecolor="black", lw=0.8))
99     ax.text(0, 0.5 * (rotor_y[0] + rotor_y[1]), "rotor core",
100            ha="center", va="center", color="white")
101
102     # ---- Dimensions stacked ABOVE the yoke (no collision) ----
103     # Row 1 (closest to yoke): w_tb/2, w_s, w_tb/2 at y_dim1
104     # Row 2 (above row 1): slot pitch tau_s at y_dim2
105     y_dim1 = 3.85
106     y_dim2 = 4.45
107
108     # w_s above the slot
109     ax.annotate("", xy=(-slot_half, y_dim1), xytext=(slot_half, y_dim1),
110                arrowprops=dict(arrowstyle="<->", lw=0.9))
111     ax.text(0, y_dim1 + 0.08, r"$w_s$", ha="center", va="bottom", fontsize=15)
112
113     # w_tb/2 on each side
114     ax.annotate("", xy=(-tau_half, y_dim1), xytext=(slot_half, y_dim1),
115                arrowprops=dict(arrowstyle="<->", lw=0.9))
116     ax.text((-tau_half - slot_half) / 2, y_dim1 + 0.08, r"$w_{tb}/2$",
117            ha="center", va="bottom", fontsize=15)
118
119     ax.annotate("", xy=(slot_half, y_dim1), xytext=(tau_half, y_dim1),
120                arrowprops=dict(arrowstyle="<->", lw=0.9))
121     ax.text((tau_half + slot_half) / 2, y_dim1 + 0.08, r"$w_{tb}/2$",
122            ha="center", va="bottom", fontsize=15)
123
124     # Slot pitch tau_s on top
125     ax.annotate("", xy=(-tau_half, y_dim2), xytext=(tau_half, y_dim2),
126                arrowprops=dict(arrowstyle="| |", lw=1.0))
127     ax.text(0, y_dim2 + 0.12, r"slot pitch $\tau_s$",
128            ha="center", va="bottom", fontsize=17)
129
130     # g and l_m dimensions on the left, outside the body
131     ax.annotate("", xy=(-4.55, airgap_y[0]), xytext=(-4.55, airgap_y[1]),
132                arrowprops=dict(arrowstyle="<->", lw=0.9))
133     ax.text(-4.75, 0.5 * (airgap_y[0] + airgap_y[1]), "$g$",
134            ha="right", va="center", fontsize=15)
135
136     ax.annotate("", xy=(-4.55, magnet_y[0]), xytext=(-4.55, magnet_y[1]),
137                arrowprops=dict(arrowstyle="<->", lw=0.9))
138     ax.text(-4.75, 0.5 * (magnet_y[0] + magnet_y[1]), "$l_m$",
139            ha="right", va="center", fontsize=15)
140
141     # R_si callout (anchored at bore line on the right)
142     ax.annotate("bore $R_{si}$", xy=(x_right, airgap_y[1]),
143                xytext=(x_right + 0.6, airgap_y[1] + 0.4),
144                arrowprops=dict(arrowstyle=">", lw=0.8),
145                ha="left", va="center", fontsize=15)
146
147     ax.set_xlim(-5.5, 6.5)
148     ax.set_ylim(-1.2, 5.0)
149     ax.set_aspect("equal")
150     ax.axis("off")
151
152     fig.tight_layout()
153     fig.savefig(OUT_DIR / "q3_geometry_sketch.png", dpi=DPI,
154                bbox_inches="tight")
155     plt.close(fig)
156
157
158     # -----
159     # 2. 1D magnetic equivalent circuit
160     # -----
161     def mec_circuit():

```

```

162 fig, ax = plt.subplots(figsize=(7.5, 4.6))
163
164 # Coordinates of the circuit nodes
165 x_m, x_g = 1.0, 4.5
166 y_bot, y_top = 0.0, 3.3
167 y_src_bot, y_src_top = 0.4, 1.0
168 y_rm_bot, y_rm_top = 1.4, 2.3
169 y_rg_bot, y_rg_top = 1.4, 2.3
170
171 # Top yoke (short)
172 ax.plot([x_m, x_g], [y_top, y_top], color="black", lw=1.4)
173 ax.text(0.5 * (x_m + x_g), y_top + 0.08,
174         r"stator yoke ( $\mathcal{R}_{Fe}\backslash\to\!0$ )",
175         ha="center", va="bottom", fontsize=15)
176
177 # Bottom yoke (short)
178 ax.plot([x_m, x_g], [y_bot, y_bot], color="black", lw=1.4)
179 ax.text(0.5 * (x_m + x_g), y_bot - 0.12,
180         r"rotor core ( $\mathcal{R}_{Fe}\backslash\to\!0$ )",
181         ha="center", va="top", fontsize=15)
182
183 # ---- Magnet branch (left) ----
184 # Wire from bottom up to MMF source
185 ax.plot([x_m, x_m], [y_bot, y_src_bot], color="black", lw=1.2)
186 # Battery: long line (+) and short line (-)
187 ax.plot([x_m - 0.30, x_m + 0.30], [y_src_bot + 0.25] * 2,
188         color="black", lw=1.6)
189 ax.plot([x_m - 0.15, x_m + 0.15], [y_src_bot + 0.45] * 2,
190         color="black", lw=1.6)
191 ax.text(x_m - 0.45, y_src_bot + 0.35, "$B_r$", ha="right",
192         va="center", fontsize=17)
193 # Wire from source up to R_m
194 ax.plot([x_m, x_m], [y_src_top, y_rm_bot], color="black", lw=1.2)
195 # R_m box
196 ax.add_patch(Rectangle((x_m - 0.30, y_rm_bot), 0.60,
197                        y_rm_top - y_rm_bot, facecolor="white",
198                        edgecolor="black", lw=1.2))
199 ax.text(x_m, 0.5 * (y_rm_bot + y_rm_top), r"$\mathcal{R}_m$",
200         ha="center", va="center")
201 ax.text(x_m + 0.45, 0.5 * (y_rm_bot + y_rm_top),
202         r"$=\dfrac{l_m}{\mu_0\mu_r A}$",
203         ha="left", va="center", fontsize=15)
204 # Wire from R_m to top yoke
205 ax.plot([x_m, x_m], [y_rm_top, y_top], color="black", lw=1.2)
206
207 # Vertical label "magnet branch"
208 ax.text(x_m - 1.7, 0.5 * (y_bot + y_top), "magnet",
209         ha="center", va="center", rotation=90, fontsize=15,
210         color="0.35")
211
212 # ---- Air-gap branch (right) ----
213 ax.plot([x_g, x_g], [y_top, y_rg_top], color="black", lw=1.2)
214 ax.add_patch(Rectangle((x_g - 0.30, y_rg_bot), 0.60,
215                        y_rg_top - y_rg_bot, facecolor="white",
216                        edgecolor="black", lw=1.2))
217 ax.text(x_g, 0.5 * (y_rg_bot + y_rg_top), r"$\mathcal{R}_g$",
218         ha="center", va="center")
219 ax.text(x_g + 0.45, 0.5 * (y_rg_bot + y_rg_top),
220         r"$=\dfrac{g}{\mu_0 A}$",
221         ha="left", va="center", fontsize=15)
222 ax.plot([x_g, x_g], [y_rg_bot, y_bot], color="black", lw=1.2)
223
224 ax.text(x_g + 2.3, 0.5 * (y_bot + y_top), "air gap",
225         ha="center", va="center", rotation=90, fontsize=15,
226         color="0.35")
227
228 # Flux arrow along the top yoke
229 arrow = FancyArrowPatch((x_m + 0.6, y_top - 0.18),
230                          (x_g - 0.6, y_top - 0.18),
231                          arrowstyle="->", mutation_scale=15,
232                          color="#1f4eaf", lw=2.0)
233 ax.add_patch(arrow)
234 ax.text(0.5 * (x_m + x_g), y_top - 0.42, r"$\Phi$",

```

```

235         ha="center", va="center", color="#1f4eaf", fontsize=18)
236
237     # Divider equation off to the side
238     ax.text(0.5 * (x_m + x_g), -0.85,
239            r"$\dfrac{B_g}{B_r}$"
240            r"=\dfrac{\mathcal{R}_m}{\mathcal{R}_m+\mathcal{R}_g}"
241            r"=\dfrac{l_m}{l_m+\mu_r g}$",
242            ha="center", va="top", fontsize=17)
243
244     ax.set_xlim(-1.2, 8.0)
245     ax.set_ylim(-1.6, 4.0)
246     ax.set_aspect("equal")
247     ax.axis("off")
248
249     fig.tight_layout()
250     fig.savefig(OUT_DIR / "q3_mec.png", dpi=DPI, bbox_inches="tight")
251     plt.close(fig)
252
253
254     # -----
255     # 3. Flux funnelling into one tooth
256     # -----
257     def flux_funnel():
258         fig, ax = plt.subplots(figsize=(7.5, 5.2))
259
260         # Geometry parameters (in arbitrary plot units)
261         tau_s_half = 3.0
262         w_tb_half = 0.95
263         y_band_bot, y_band_top = 2.4, 2.9
264         y_funnel_bot, y_funnel_top = 0.6, 2.4
265         y_tooth_bot, y_tooth_top = -1.4, 0.6
266
267         # Air-gap band (slot pitch wide, B_g,avg)
268         ax.add_patch(Rectangle((-tau_s_half, y_band_bot),
269                                2 * tau_s_half,
270                                y_band_top - y_band_bot,
271                                facecolor="#d6e7ff",
272                                edgecolor="black", lw=0.8))
273
274         # Slot pitch brace + label
275         ax.annotate("", xy=(-tau_s_half, y_band_top + 0.4),
276                    xytext=(tau_s_half, y_band_top + 0.4),
277                    arrowprops=dict(arrowstyle="|-|", lw=1.0))
278         ax.text(0, y_band_top + 0.55,
279                r"one slot pitch $\tau_s$ (flux density $B_{g,\mathrm{avg}}$)",
280                ha="center", va="bottom")
281
282         # Downward flux arrows over the full slot pitch
283         for x in [-2.7, -2.1, -1.5, -0.9, -0.3, 0.3, 0.9, 1.5, 2.1, 2.7]:
284             arrow = FancyArrowPatch((x, y_band_top - 0.05),
285                                    (x, y_band_bot + 0.05),
286                                    arrowstyle="->", mutation_scale=10,
287                                    color="#1f4eaf", lw=1.4)
288
289             ax.add_patch(arrow)
290
291         # Funnel (gray flux-collecting region)
292         funnel = Polygon([(-tau_s_half, y_band_bot),
293                           (-w_tb_half, y_funnel_bot),
294                           (w_tb_half, y_funnel_bot),
295                           (tau_s_half, y_band_bot)],
296                           closed=True, facecolor="0.88",
297                           edgecolor="black", lw=0.9)
298         ax.add_patch(funnel)
299
300         # Tooth body
301         ax.add_patch(Rectangle((-w_tb_half, y_tooth_bot),
302                                2 * w_tb_half,
303                                y_tooth_top - y_tooth_bot,
304                                facecolor="0.82",
305                                edgecolor="black", lw=0.9))
306
307         # Flux arrows inside the tooth (thicker, fewer) -- stop above the label
308         arrow_tip_y = -0.3

```

```

308     for x in [-0.5, -0.15, 0.15, 0.5]:
309         arrow = FancyArrowPatch((x, y_funnel_bot - 0.05),
310                                 (x, arrow_tip_y),
311                                 arrowstyle="->", mutation_scale=14,
312                                 color="#1f4eaf", lw=2.0)
313         ax.add_patch(arrow)
314
315     # "tooth (B_t)" label placed BELOW the arrows
316     ax.text(0, y_tooth_bot + 0.35, r"tooth ($B_t$)",
317            ha="center", va="center", fontsize=17)
318
319     # w_tb dimension below the tooth
320     y_w = y_tooth_bot - 0.25
321     ax.annotate("", xy=(-w_tb_half, y_w),
322                xytext=(w_tb_half, y_w),
323                arrowprops=dict(arrowstyle="<->", lw=1.0))
324     ax.text(0, y_w - 0.15, r"$w_{tb}$", ha="center", va="top")
325
326     # Side labels "slot" on either side of the funnel
327     ax.text(-2.0, 1.4, "slot", ha="center", va="center", fontsize=15,
328            color="0.35")
329     ax.text(2.0, 1.4, "slot", ha="center", va="center", fontsize=15,
330            color="0.35")
331
332     # Conservation equation
333     ax.text(0, -2.4,
334            r"$B_{g,\mathrm{avg}}\cdot\tau_s \;=\; B_t\cdot w_{tb}$"
335            r"      (flux conservation)",
336            ha="center", va="center", fontsize=17)
337
338     ax.set_xlim(-4.0, 4.0)
339     ax.set_ylim(-3.0, 4.0)
340     ax.set_aspect("equal")
341     ax.axis("off")
342
343     fig.tight_layout()
344     fig.savefig(OUT_DIR / "q3_flux_funnel.png", dpi=DPI,
345                bbox_inches="tight")
346     plt.close(fig)
347
348
349 if __name__ == "__main__":
350     geometry_sketch()
351     mec_circuit()
352     flux_funnel()
353     print("Wrote q3_geometry_sketch.png, q3_mec.png, q3_flux_funnel.png")

```